REWEIGHTED ADMM-BASED ROBUST DOA ESTIMATION FOR MULTIPLE SHIPS USING VSA

Thesis submitted to the SASTRA Deemed to be University
in partial fulfillment of the requirements
for the award of the degree of

B. Tech. Electronics and Communication Engineering

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This is to certify that the thesis titled "**Reweighted ADMM-Based Robust DOA Estimation For Multiple Ships Using VSA**" submitted in partial fulfillment of the requirements for the award of the degree of B. Tech. Electronics and Communication Engineering to the SASTRA Deemed to be University, is a bona-fide record of the work done by **Ms. Afreen A** (Reg. No. 126004008), **Mr. Deepakraja C**(Reg. No. 126004063), **Ms. Priyadharshini J**(Reg. No. 126004206) during the Eighth semester of the academic year 2025-26, in the **School of Electrical & Electronics Engineering**, under my supervision. This thesis has not formed the basis for the award of any degree, diploma, associateship, fellowship or other similar title to any candidate of any University.

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Date :

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Examiner 1 **Examiner 2**

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Declaration

We declare that the thesis titled "**REWEIGHTED ADMM-BASED ROBUST DOA ESTIMATION FOR MULTIPLE SHIPS USING VSA**" submitted by us is an original work done by us under the guidance of **Dr. Priya SBM, Senior Assistant Professor, School of Electrical and Electronics Engineering, SASTRA Deemed to be University** during the Eighth semester of the academic year **2025-26**, in the **School of Electrical and Electronics Engineering**. The work is original and wherever we have used materials from other sources, we have given due credit and cited them in the text of the thesis. This thesis has not formed the basis for the award of any degree, diploma, associate-ship, fellowship or other similar title to any candidate of any University.

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Abstract

Accurate estimation of the Direction of Arrival (DOA) is essential for monitoring ship activity in harbor and coastal environments, where multiple acoustic sources often overlap. This work adopts a passive sensing approach using a Circular Vector Sensor Array (VSA), which captures both acoustic pressure and particle velocity to improve spatial resolution. The DOA estimation problem is formulated as a sparse reconstruction task and solved using the Alternating Direction Method of Multipliers (ADMM) optimization framework with a reweighted $\ell_1$strategy. To account for practical challenges such as sensor misalignment, the model incorporates Non-Orthogonal Deviation (NOD), while robustness against impulsive noise is enhanced through nonlinear preprocessing and adaptive weighting. The proposed method is further compared with the Multiple Signal Classification (MUSIC) algorithm to highlight improvements in resolution and noise tolerance. Simulation results demonstrate that the approach achieves accurate DOA estimation in multi-source and noisy conditions while maintaining stable computational performance.

Specific Contribution:

Used the MUSIC algorithm for DOA estimation by separating signal and noise subspaces, and applied OMMC calibration to correct sensor misalignment (NOD) for improved accuracy.

Specific Learning:

Understood how the MUSIC algorithm estimates DOA using subspace separation and how OMMC calibration improves its accuracy by correcting sensor imperfections (NOD).

Keywords: *Direction of Arrival (DOA), Vector Sensor Array (VSA), Alternating Direction Method of Multipliers (ADMM) Optimization, Non-Orthogonal Deviation (NOD), Multiple Signal Classification (MUSIC).*

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Abstract

Estimating the Direction of Arrival (DOA) of underwater acoustic sources is challenging in harbour and coastal environments due to overlapping signals from multiple ships. A passive sensing framework using a Circular Vector Sensor Array (VSA) is adopted to capture both acoustic pressure and particle velocity, enabling improved directional resolution. The DOA estimation problem is formulated as a sparse recovery task and solved using an Alternating Direction Method of Multipliers (ADMM) optimization approach with reweighted $\ell_1$regularization. To reflect practical conditions, Non-Orthogonal Deviation (NOD) in sensor orientation is incorporated into the model. Robustness to impulsive and non-Gaussian noise is further improved through nonlinear preprocessing and adaptive weighting. The method is evaluated through simulations and compared with the Multiple Signal Classification (MUSIC) algorithm. Results demonstrate improved resolution and reliable DOA estimation in multi-source and noisy scenarios, while maintaining stable computational performance.

Specific Contribution:

Developed the DOA estimation pipeline using VSA, including signal modeling, steering vector design and spectrum analysis. Contributed to noise handling and evaluated performance under interference conditions.

Specific Learning:

Developed understanding of array signal processing concepts including subspace methods, noise effects on DOA estimation accuracy, and how VSA enhance directional sensing compared to conventional arrays.

Keywords: *Direction of Arrival (DOA), Vector Sensor Array (VSA), Alternating Direction Method of Multipliers (ADMM) Optimization, Non-Orthogonal Deviation (NOD), Multiple Signal Classification (MUSIC).*

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Abstract

Monitoring ship movement in harbour and coastal regions is crucial for maritime safety. Ships generate underwater acoustic noise, and in multi-ship scenarios, overlapping signals make Direction of Arrival (DOA) estimation challenging. This project uses a passive approach with a Circular Vector Sensor Array (VSA), which captures acoustic pressure and particle velocity for improved directional sensing. DOA estimation is formulated as a sparse recovery problem using reweighted L1 minimization and solved using the Alternating Direction Method of Multipliers (ADMM). To enhance robustness under impulsive noise, nonlinear preprocessing and adaptive reweighting are applied. MATLAB simulations demonstrate accurate and high-resolution DOA estimation in multi-target and noisy conditions.

Specific Contribution:

Implemented a reweighted L1-based ADMM algorithm in MATLAB for sparse DOA estimation using a VSA and validated its performance under multi-target and noisy conditions.

Specific Learning:

Gained understanding of Direction of Arrival (DOA) estimation techniques and challenges in multi-source environments, along with the working principles and advantages of VSA.

Keywords: *Direction of Arrival (DOA), Vector Sensor Array (VSA), Alternating Direction Method of Multipliers (ADMM) Optimization, Non-Orthogonal Deviation (NOD), Multiple Signal Classification (MUSIC).*

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Table of Contents

Title	Page No.
Bona-fide Certificate	ii
Declaration	iii
Acknowledgements	v
Abstract	vi
List of Figures	vii
List of Tables	viii
Abbreviations	ix
Notations	x
CHAPTER 1: INTRODUCTION	1
1.1 Integrated Sensing and Communication Systems	1
1.2 Direction of Arrival (DOA) Estimation	1
1.3 Vector Sensor Arrays (VSA)	2
1.4 Challenges in DOA Estimation under Noise	2
1.5 Sparse Signal Processing for DOA Estimation	3
1.6 ADMM Optimization Techniques	4
CHAPTER 2: LITERATURE SURVEY	5
2.1 Literature review	5
2.2 Comparative Analysis of Existing and Proposed Method	7
CHAPTER 3: OBJECTIVE	8
3.1 Problem Definition	8
3.2 Objectives of the Work	9
3.3 Scope of the Project	10

CHAPTER 4: METHODOLOGY 11
4.1 Circular VSA-Based System Model Setup 11
4.2 Sparse DOA Formulation Using Reweighted L1 Minimization 12
4.3 Robust DOA Estimation Using ADMM Optimization 14
4.4 Robust Reweighted ADMM Algorithm 16
4.5 Spatial Spectrum Generation and DOA Estimation 18

CHAPTER 5: RESULTS AND DISCUSSION 19
5.1 Simulation Results under Different Scenarios 19
5.1.1 Baseline Results of MUSIC Algorithm with OMMC 20
5.1.2 Simulation Results of MUSIC with OMMC under Noise and Interference 22
5.1.3 Proposed Method: DOA Estimation using Reweighted ADMM 26
5.2 Performance Analysis 30
5.2.1 Performance under Existing Method (MUSIC with OMMC) 30
5.2.2 Performance of MUSIC with OMMC under Noise and Interference 35
5.2.3 Performance of Reweighted ADMM-Based Method 36
5.2.3.1 Effect of SNR on Estimation Accuracy 37
5.2.3.2 Influence of Array Size and Number of Snapshots 39
5.2.3.3 Effect of NOD on Estimation Bias 41
5.2.3.4 Resolution Performance of Closely Spaced Targets 42
5.2.3.5 Computational Efficiency Analysis 44
5.3Comparative Analysis 45

CHAPTER 6: CONCLUSION AND FUTURE SCOPE 48
6.1 Conclusion 48
6.2 Future Work 49

CHAPTER 7: REFERENCES 51

CHAPTER 8: APPENDICES 53
8.1 Similarity Check Report 53

List of Figures

Fig. No.	Title	Page No.
1	Acoustic Vector Sensor Array for DOA Estimation	2
2	Illustration of Underwater Acoustic Signal Propagation	3
3	DOA Estimation using MUSIC under Ideal Conditions	21
4	MUSIC Spatial Spectrum (Calibrated vs Uncalibrated)	22
5	Estimated DOA Values under Non-Ideal Conditions	24
6	MUSIC-OMMC Spatial Spectrum under Noise	25
7	DOA Estimation using Reweighted ADMM Method	28
8	Spatial Spectrum using Reweighted ADMM	29
9	RMSE vs SNR (MUSIC + OMMC)	31
10	CRLB vs SNR under Different NOD	32
11	RMSE versus DOA for Different NOD Levels	33
12	Bias vs DOA for Different NOD Levels	34
13	MUSIC-OMMC Spectrum under Interference	36
14	RMSE vs SNR (Reweighted ADMM)	37
15	CRLB vs SNR (Non-Ideal Conditions)	38
16	CRLB vs Number of Sensors	39
17	CRLB vs Number of Snapshots	40
18	Bias vs Number of Sensors	41
19	Resolution of Closely Spaced Targets	42
20	Probability of Resolution vs SNR	43
21	CPU Time Comparison	44

List of Tables

Table No.	Title	Page No.
1	Comparative Analysis of DOA Techniques	7
2	Simulation Parameters (Ideal Case)	20
3	Simulation Parameters (Non-Ideal Case – MUSIC)	23
4	Simulation Parameters (Proposed Method)	27
5	Complexity Comparison (MUSIC vs ADMM)	46
6	Performance Comparison	47

Abbreviations

DOA	Direction of Arrival
VSA	Vector Sensor Array
ADMM	Alternating Direction Method of Multipliers
LASSO	Least Absolute Shrinkage and Selection Operator
MUSIC	Multiple Signal Classification
NOD	Non-Orthogonal Deviation
SNR	Signal-to-Noise Ratio
RMSE	Root Mean Square Error
CRLB	Cramér–Rao Lower Bound
ANM	Atomic Norm Minimization
R-ANM	Reweighted Atomic Norm Minimization
OMMC	Orthogonal Multi-Modal Calibration
FFT	Fast Fourier Transform
SVD	Singular Value Decomposition
ULA	Uniform Linear Array
UCA	Uniform Circular Array
PDF	Probability Density Function
SPL	Sound Pressure Level
SL	Source Level
AIS	Automatic Identification System
CPA	Closest Point of Approach
FOC	Fourth-Order Cumulants
RIS	Reconfigurable Intelligent Surface

Notations

M	Number of sensors in the array
K	Number of signal sources
Ns	Number of snapshots
θ	Direction of Arrival (DOA) angle
$\boldsymbol{\theta_{true}}$	True DOA values
$\boldsymbol{\theta_{grid}}$	Discretized angular grid
Ng	Number of grid points
r	Received signal vector
x	Sparse signal vector
$\mathbf{x_{true}}$	True signal vector
$\boldsymbol{x_est}$	Estimated signal vector
n	Noise vector
G	Steering matrix
$\boldsymbol{G_{ideal}}$	Ideal steering matrix
Ψ	Non-Orthogonal Deviation (NOD) matrix
R	Radius of circular array
φm	Angular position of sensor
z	Auxiliary variable (ADMM)
u	Dual variable (ADMM)
ρ	ADMM penalty parameter
λ	Regularization parameter
$\boldsymbol{W_i}$	Reweighting coefficient
ε	Small constant (numerical stability)
$(\cdot)^{H}$	Hermitian transpose
$\|\cdot\|_1$	L1 norm
$\|\cdot\|_2$	L2 norm
P(θ)	Spatial spectrum

$\hat{\theta}$	Estimated DOA
α	Stability parameter (alpha-stable noise)
γ	Scale parameter (noise)
SPL	Sound Pressure Level
SL	Source Level
Ra	Range of source

CHAPTER 1
INTRODUCTION

1.1 Integrated Sensing and Communication Systems

In recent years, there has been a growing need to combine sensing and communication systems to improve efficiency and performance. Integrated Sensing and Communication (ISAC) systems are designed to perform both tasks simultaneously using the same hardware and signal resources. This integration is especially useful in applications such as radar, wireless communication, and underwater monitoring.

In underwater environments, ISAC systems help in detecting objects like ships while also enabling communication between devices. Instead of using separate systems for sensing and communication, a unified approach reduces complexity, saves energy, and improves overall system performance. This makes ISAC an important concept in modern signal processing applications.

1.2 Direction of Arrival (DOA) Estimation

Direction of Arrival (DOA) estimation is a technique used to determine the direction from which a signal is received by an array of sensors. It is widely used in applications such as sonar, radar, wireless communication, and surveillance systems.

In underwater scenarios, ships generate acoustic signals that travel through water and are captured by sensors. By analyzing these signals, it is possible to estimate the direction of the source. Accurate DOA estimation helps in tracking multiple ships, avoiding collisions, and improving navigation systems.

However, when multiple signals arrive simultaneously or when noise is present, identifying the correct direction becomes more challenging. Therefore, advanced signal processing techniques are required to achieve reliable DOA estimation.

1.3 Vector Sensor Arrays (VSA)

Vector Sensor Arrays (VSA) are an advanced type of sensor system used for DOA estimation. Unlike traditional scalar sensors that measure only the pressure of a signal, VSAs measure both acoustic pressure and particle velocity components.

This additional information provides more insight into the direction of incoming signals, allowing for better accuracy and resolution. VSAs are particularly useful in underwater environments where signals can reflect, scatter, and interfere with each other.

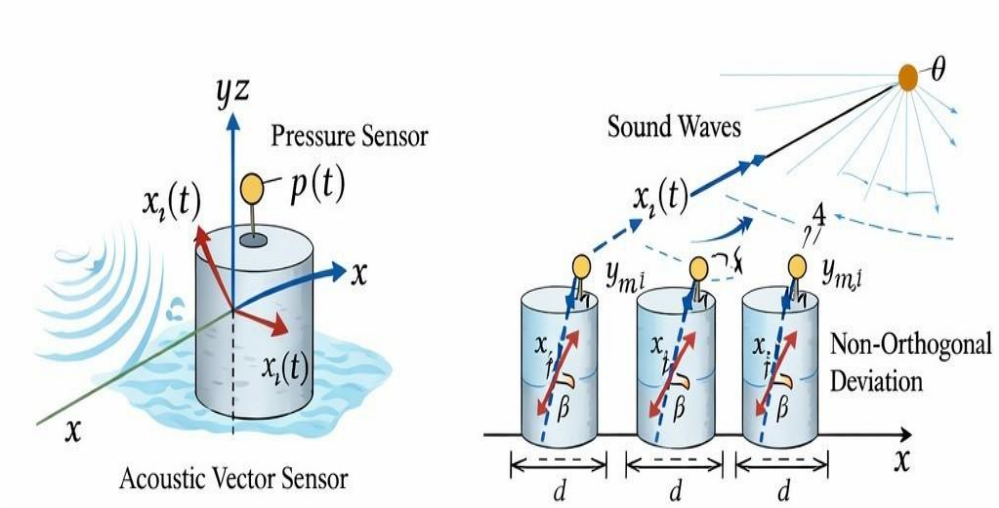


Figure 1: Acoustic Vector Sensor Array for Direction of Arrival Estimation

By using VSAs, it becomes possible to detect multiple sources more effectively and improve the overall performance of DOA estimation systems.

1.4 Challenges in DOA Estimation under Noise

In real-world environments, DOA estimation is affected by several challenges, especially due to noise and interference. Noise can be of different types, including Gaussian noise and impulsive (non-Gaussian) noise, which can distort the received signals.

Additionally, interference from other sources, such as nearby ships or environmental disturbances, can make it difficult to distinguish between signals. When multiple signals overlap, it becomes even more challenging to identify their directions accurately.

Another important issue is sensor imperfections, such as misalignment or calibration errors, which can lead to incorrect estimations. These challenges reduce the effectiveness of traditional DOA estimation methods and require more robust approaches.

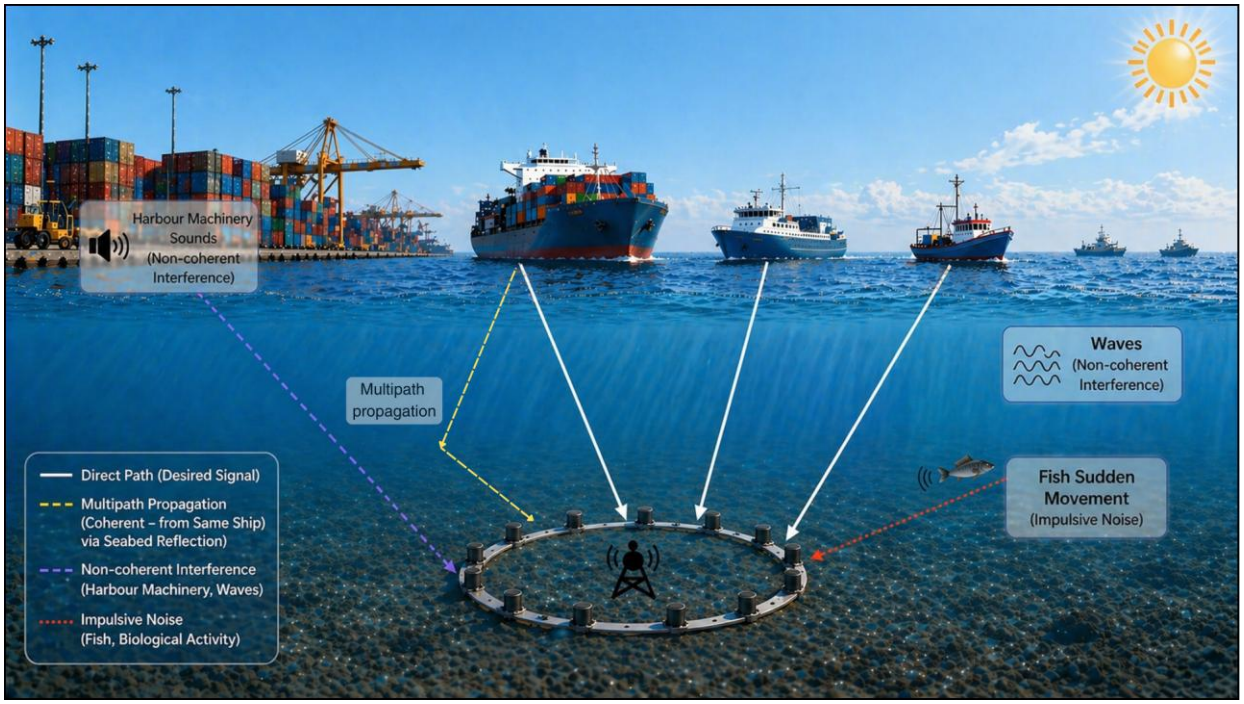


Figure 2: Illustration of Underwater Acoustic Signal Propagation and Interference in VSA-Based DOA Estimation

1.5 Sparse Signal Processing for DOA Estimation

Sparse signal processing is an advanced approach used to improve DOA estimation in complex environments. The main idea behind this method is that only a few directions contain actual signal sources, while the rest contain noise or no signal.

By treating DOA estimation as a sparse problem, it becomes possible to identify only the important directions where signals are present. This approach improves resolution and allows the system to detect multiple sources even when they are closely spaced.

Sparse methods are particularly useful in handling interference and noise, making them more reliable than traditional techniques in real-world conditions.

1.6 ADMM Optimization Techniques

The Alternating Direction Method of Multipliers (ADMM) is an optimization technique used to solve complex problems in a simple and efficient manner. It works by breaking a large problem into smaller subproblems and solving them iteratively.

In DOA estimation, ADMM is used to solve sparse optimization problems by updating variables step by step. It ensures that the solution gradually improves with each iteration until it converges to an optimal result.

One of the main advantages of ADMM is its ability to handle large-scale problems and constraints effectively. This makes it suitable for real-time applications and complex signal processing tasks such as multi-target DOA estimation**.**

CHAPTER 2
LITERATURE SURVEY

1.1 Literature Review

1) Autonomous Detection, Localization and Tracking of Ships using Vector Sensor Array – Shweta Lokhande et al.

This paper presents a real-time underwater system that detects and tracks ships using Vector Sensor Arrays (VSA). It uses the MUSIC algorithm with SVD to estimate DOA and track ship movement continuously. The results show that VSA provides accurate localization and reliable performance for coastal surveillance applications.

2) Direction Finding using Acoustic Vector Sensor Array with Fluctuating Misorientation – Weidong Wang et al.

This paper addresses the problem of sensor misalignment (orientation errors) in vector sensor arrays. It proposes a Covariance Matrix Focusing Fitting (CMFF) method to correct these errors and improve DOA estimation. The method enhances accuracy even when sensors are affected by environmental disturbances like currents and motion.

3)Direction Finding using Acoustic Vector Sensor Array with Non-Orthogonal Deviation – Weidong Wang et al.

This paper studies the effect of sensor misalignment (non-orthogonal deviation) on DOA estimation in acoustic vector sensor arrays. It proposes an Optimal Modification Matrix Construction (OMMC) method to correct these deviations using an auxiliary reference source. The method improves estimation accuracy and robustness, especially in challenging conditions with noise and limited data.

4)Efficient DOA Estimation Method for Extended-Aperture Moving Coprime Arrays Based on ANM-ADMM – Jiawei Wang et al.

This paper introduces an efficient DOA estimation method using Atomic Norm Minimization (ANM) combined with ADMM. It improves estimation accuracy by reconstructing missing data in virtual arrays and enhancing array aperture. The method reduces computational complexity while achieving better performance in scenarios with multiple targets and noise.

5) Passive Direction Finding using Acoustic Vector Sensors with Fourth-Order Cumulants – Ting Shu, Jin He

This paper proposes a direction-finding algorithm using fourth-order cumulants (FOC) for better noise handling. It enables accurate DOA estimation even in correlated noise and underdetermined conditions. The method improves detection capability by using virtual sensors and coarray techniques.

6) High-Accuracy DOA Estimation in RIS-Assisted Systems under Multiple Interference Sources – S. Priyadharshini, S.B.M Priya.

This paper focuses on improving Direction of Arrival (DOA) estimation using Reconfigurable Intelligent Surfaces (RIS) in environments with multiple interference sources. It proposes an ANM-ADMM-based method to reduce computational complexity while maintaining high accuracy. The study highlights that considering both direct and scattered interference significantly improves real-world performance.

7) Underwater Target Localization using DOA and ML – Wang Biao et al.

This paper focuses on underwater target localization using a network of acoustic sensor arrays. It utilizes Direction of Arrival (DOA) estimates from multiple sensor nodes and applies a Maximum Likelihood (ML) method to determine the target position. The approach models the statistical behavior of DOA measurements and combines them to improve localization accuracy. The results show that the method provides reliable performance and better accuracy compared to traditional least square methods in underwater environments.

2.2 Comparative Analysis of Existing DOA Estimation Techniques and Proposed Method

Ref	Technique	Array Type	Interference Handling	Handles Unknown Number of Targets	Algorithm Type
[3]	Autonomous Detection using Vector Sensor Array	VSA	✓	✓	Signal Processing
[4]	OMMC-based Robust DOA Estimation under NOD	AVSA	✓	✗	Calibration-assisted + Subspace (MUSIC)
[7]	ANM-ADMM for DOA Estimation in Coprime Arrays	Coprime Array	✓	✓	ANM-ADMM (Convex Optimization)
[10]	DOA with Fluctuating Misorientation	VSA	✗	✗	Calibration-based
[11]	Passive DOA using Vector Sensors (Cumulants)	VSA	✓	✓	Higher-order Statistics
Proposed	Reweighted ADMM	Circular VSA	✓ (Coherent & Non-Coherent)	✓	Reweighted ADMM

Table 1: Comparative Analysis of Existing DOA Estimation Techniques and Proposed Method

CHAPTER 3
OBJECTIVES

3.1 Problem Definition

Underwater acoustic environments, particularly in coastal and harbour regions, are inherently complex due to the presence of multiple simultaneously operating vessels and marine activities. These sources emit acoustic signals that propagate through water and are received collectively at sensor arrays, resulting in overlapping signal observations. This overlap makes accurate estimation of the Direction of Arrival (DOA) significantly more difficult.

Signal propagation in underwater environments is affected by several physical phenomena. Reflections from the sea surface and seabed introduce multipath effects, causing distortion and time spreading of signals. In addition, interference plays a critical role in degrading signal quality. Coherent interference arises when sources generate correlated signals, such as similar mechanical components in ships, whereas non-coherent interference originates from unrelated sources like small boats, environmental noise, and wave activity. Furthermore, impulsive noise characterized by sudden high energy disturbances and non-Gaussian behavior poses a major challenge for conventional estimation techniques.

Another practical limitation is related to sensor imperfections. In real-world deployments, vector sensors often deviate from ideal orthogonal alignment, leading to Non-Orthogonal Deviation (NOD). This misalignment affects the accuracy of the steering vector and introduces systematic bias in DOA estimation.

Traditional subspace-based methods, such as MUSIC, are developed under ideal assumptions including Gaussian noise, uncorrelated sources, and prior knowledge of the number of targets. However, these assumptions rarely hold in practical underwater scenarios. As a result, such methods often exhibit degraded performance, including incorrect peak detection, reduced resolution, and inaccurate localization.

To overcome these limitations, this work reformulates the DOA estimation problem within a sparse signal reconstruction framework. A robust optimization strategy based on reweighted L1 minimization, solved using the Alternating Direction Method of Multipliers (ADMM), is proposed. This approach improves estimation reliability and enables effective multi-target detection under realistic and challenging underwater conditions.

3.2 Objectives of the Work

The main goal of this project is to develop a robust and efficient framework for estimating the Direction of Arrival (DOA) of multiple acoustic sources in underwater environments using Vector Sensor Arrays and sparse optimization techniques. the specific objectives are as follows:

1. To design and implement a MATLAB-based circular Vector Sensor Array (VSA) model that captures both acoustic pressure and particle-velocity components for improved directional analysis.
2. To model realistic underwater acoustic environments by incorporating:
    - Coherent interference
    - Non-coherent interference
    - Impulsive noise using alpha-stable distributions
3. To model sensor imperfections through a Non-Orthogonal Deviation (NOD) framework and evaluate its impact on estimation accuracy.
4. To implement a conventional DOA estimation technique (MUSIC) and analyze its performance under both ideal and non-ideal conditions for baseline comparison.
5. To reformulate the DOA estimation problem as a sparse reconstruction problem using a discretized angular domain and L1-norm regularization.
6. To develop an advanced DOA estimation method based on reweighted L1 minimization using the ADMM optimization framework.
7. To enable automatic detection of multiple targets without requiring prior knowledge of the number of sources.

8. To improve estimation performance by incorporating:
    - Iterative reweighting
    - Adaptive regularization
    - Noise suppression mechanisms
9. To evaluate the system using performance metrics such as:
    - Root Mean Square Error (RMSE)
    - Cramér–Rao Lower Bound (CRLB)
    - Estimation bias
    - Resolution capability
    - Detection probability under varying Signal-to-Noise Ratio (SNR) and snapshot conditions

3.3 Scope of the Project

This project focuses on the development and evaluation of a multi-target DOA estimation framework for underwater acoustic environments using simulation-based analysis. A circular Vector Sensor Array (VSA) is employed to provide full angular coverage and improved spatial resolution for direction estimation.

The signal model incorporates realistic environmental effects, including multipath propagation, coherent and non-coherent interference, impulsive noise, and sensor imperfections such as Non-Orthogonal Deviation. The proposed approach formulates DOA estimation as a sparse optimization problem and employs a reweighted L1 minimization strategy solved using ADMM to achieve high-resolution estimation without prior knowledge of the number of sources. Performance evaluation is carried out using MATLAB simulations, where the proposed method is compared with conventional approaches to demonstrate improvements in accuracy, robustness, and resolution under challenging conditions.

It should be noted that the current work is limited to simulation-based analysis and does not include hardware implementation. Due to the iterative nature of the optimization algorithm, computational complexity is relatively high, which may pose challenges for real-time deployment.

CHAPTER 4
METHODOLOGY

4.1 Circular VSA-Based System Model Setup

In this work, a Circular Vector Sensor Array (VSA) is employed to estimate the Direction of Arrival (DOA) of multiple signal sources in a complex noise environment. Compared to conventional Uniform Linear Arrays (ULA), the circular configuration provides complete 360° angular coverage and improved spatial resolution due to its symmetric geometry. The array consists of Msensor elements uniformly distributed along a circular structure of radius R. The angular position of the m^{th}sensor is defined as

$$\phi_m = \frac{2\pi(m-1)}{M}, m = 1,2,\dots,M \text{ where } \phi_m \in \mathbb{R}.$$

This uniform placement ensures that the array captures signals arriving from all directions without directional bias. For a signal arriving from direction θ, the response at each sensor is modeled using the steering vector. The response of the m^{th}sensor is given by

$$a_m(\theta) = e^{j2\pi\, \mathrm{Rcos}(\theta-\phi_m)} \text{ where } a_m(\theta) \in \mathbb{C}.$$

The complete steering vector is expressed as

$$a(\theta) = [a_1(\theta), a_2(\theta), \dots, a_M(\theta)]^T \in \mathbb{C}^{M\times 1}$$

By evaluating this steering vector over a discretized angular grid defined as

$$\theta = \{0°, 1°, 2°, \dots, 359°\}$$

the number of grid points is denoted as N_g.

The steering matrix is constructed as

$$G = [a(\theta_1), a(\theta_2), \dots, a(\theta_{N_g})] \in \mathbb{C}^{3M\times N_g}$$

The received signal model is given by

$$R = GX + N$$

where

$R \in \mathbb{C}^{3M \times N_s}$ denotes the received signal matrix,

$G \in \mathbb{C}^{3M \times N_g}$ is the steering matrix,

$X \in \mathbb{C}^{N_g \times N_s}$ represents the source signal matrix, and

$N \in \mathbb{C}^{3M \times N_s}$ is the additive noise matrix.

where r $\in \mathbb{C}^{M \times 1}$ represents the received signal vector, Gis the steering matrix, x $\in \mathbb{C}^{N_g \times 1}$ is the sparse signal vector indicating the presence of sources at specific angles, and n $\in \mathbb{C}^{M \times 1}$ denotes the noise component.

In practical scenarios, sensor arrays are affected by imperfections such as misalignment, gain variations, and mutual coupling effects. To account for these non-idealities, a Non-Orthogonal Deviation (NOD) model is incorporated. The ideal steering matrix is modified using a transformation matrix Ψ, resulting in an effective steering matrix given by:

$$\mathrm{G}_{eff} = \Psi \mathrm{G}$$

This modification improves the accuracy of the model under realistic conditions.
Furthermore, the noise is modelled using an alpha-stable distribution to capture impulsive and heavy-tailed characteristics commonly observed in practical environments. This makes the estimation problem more challenging and necessitates robust processing techniques.

Based on the above system model, the objective is to estimate the sparse signal vector xfrom the observed signal r. The non-zero entries of xcorrespond to the true directions of arrival of the signal sources, which are subsequently used for DOA estimation.

4.2 Sparse DOA Formulation Using Reweighted L1 Minimization

The Direction of Arrival (DOA) estimation problem can be formulated as a sparse signal recovery problem in the angular domain. In practical scenarios, only a limited number of signal sources are present, whereas the angular search space is large. This results in a sparse representation, where only a few components of the unknown signal vector are non-zero.

To exploit this sparsity, the angular domain is discretized into a finite grid defined as

$$\theta = \{0°, 1°, 2°, \dots, 359°\}$$

where the total number of grid points is denoted by N_g.

Each grid point represents a potential direction of arrival. Since only a few directions correspond to actual sources, the signal vector exhibits sparsity.

Using the effective steering matrix derived in Section 4.1, the received signal model is expressed as

$$r = G_{\text{eff}} x + n$$

where

$r \in \mathbb{C}^{3M \times 1}$ denotes the received signal vector,

$G_{\text{eff}} \in \mathbb{C}^{3M \times N_g}$ is the effective steering matrix,

$x \in \mathbb{C}^{N_g \times 1}$ is a sparse vector whose non-zero entries correspond to the true DOA locations, and

$n \in \mathbb{C}^{3M \times 1}$ represents additive noise.

The estimation of x is formulated as the following optimization problem:

$$\min_{x} \quad \| r - G_{\text{eff}} x \|_2^2 + \lambda \sum_{i=1}^{N_g} w_i \, | \, x_i \, |$$

where $\| r - G_{\text{eff}} x \|_2^2$ is the data fidelity term that ensures consistency with the observed signal, and the second term enforces sparsity through weighted ℓ_1-regularization. The parameter λcontrols the trade-off between reconstruction accuracy and sparsity.

To further enhance sparsity and improve resolution, a reweighting strategy is employed. Unlike standard ℓ_1 minimization, which applies uniform penalization, the reweighted approach adaptively assigns different weights to different coefficients, thereby improving the detection of closely spaced sources.

The weights are updated iteratively based on the current estimate as:

$$w_i = \frac{1}{\text{mean}(| \, z_i \, |) + \delta}$$

where $w_i \in \mathbb{R}$, z_i denotes the intermediate estimate of the sparse vector, and $\delta > 0$ is a small positive constant introduced to avoid numerical instability.

This formulation assigns smaller penalties to dominant components and larger penalties to weaker components, effectively suppressing noise and enhancing significant DOA peaks.

This formulation is based on a computationally efficient grid-based sparse recovery framework using reweighted ℓ_1 minimization. Compared to traditional high-complexity approaches, this method provides a practical balance between computational efficiency and estimation accuracy.

Thus, the DOA estimation problem is transformed into a sparse optimization problem, where the goal is to recover the sparse vector $x \in \mathbb{C}^{N_g \times 1}$ from the observed signal $r \in \mathbb{C}^{3M \times 1}$. The solution obtained from this formulation is further refined using optimization techniques, as discussed in the following section.

4.3 Robust DOA Estimation Using ADMM Optimization

To efficiently solve the sparse DOA estimation problem formulated in Section 4.2, the Alternating Direction Method of Multipliers (ADMM) is employed. ADMM is an iterative optimization technique that decomposes a complex problem into simpler subproblems, enabling efficient computation while maintaining convergence.

The original optimization problem is given by

$$\min_{x} \quad \| r - G_{\text{eff}} x \|_2^2 + \lambda \sum_{i=1}^{N_g} w_i \mid x_i \mid$$

where $r \in \mathbb{C}^{3M \times 1}$, $G_{\text{eff}} \in \mathbb{C}^{3M \times N_g}$, and $x \in \mathbb{C}^{N_g \times 1}$.

To facilitate ADMM-based optimization, an auxiliary variable $z \in \mathbb{C}^{N_g \times 1}$ is introduced, leading to the reformulated problem:

$$\min_{x,z} \quad \| r - G_{\text{eff}} x \|_2^2 + \lambda \| z \|_1 \text{ subject to} x = z$$

This reformulation separates the data fidelity term and the sparsity constraint, allowing them to be handled independently.

The corresponding augmented Lagrangian is expressed as:

$$L(x,z,u) = \| r - G_{\text{eff}} x \|_2^2 + \lambda \| z \|_1 + \frac{\rho}{2} \| x - z + u \|_2^2$$

where $u \in \mathbb{C}^{N_g \times 1}$ is the scaled dual variable and $\rho > 0$ is a penalty parameter that controls convergence.

The ADMM algorithm proceeds by iteratively updating the variables x, z, and u as follows:

x-update

The variable $x \in \mathbb{C}^{N_g \times 1}$is updated by solving a regularized least squares problem:

$$x^{(k+1)} = (G_{\text{eff}}^H G_{\text{eff}} + \rho I + \gamma L^T L)^{-1} \left(G_{\text{eff}}^H r + \rho \left(z^{(k)} - u^{(k)} \right) \right)$$

where

$I \in \mathbb{R}^{N_g \times N_g}$ is the identity matrix and

$L \in \mathbb{R}^{(N_g - 1) \times N_g}$ is a difference operator matrix used for smoothness regularization.

This step ensures that the estimated signal remains consistent with the observed measurements while incorporating regularization for stability.

z-update (Sparsity Enforcement)

The variable $z \in \mathbb{C}^{N_g \times 1}$ is updated using a soft-thresholding operator:

$$z^{(k+1)} = \text{sign}\left(x^{(k+1)} + u^{(k)} \right) \cdot \max\left(| x^{(k+1)} + u^{(k)} | - \lambda w, 0 \right)$$

where $w \in \mathbb{R}^{N_g \times 1}$ is the weight vector.

This operation enforces sparsity by shrinking small coefficients toward zero, thereby retaining only significant components corresponding to true DOA values.

u-update (Dual Variable Update)

The dual variable $u \in \mathbb{C}^{N_g \times 1}$ is updated as:

$$u^{(k+1)} = u^{(k)} + x^{(k+1)} - z^{(k+1)}$$

This step ensures consistency between the variables $x$and $z$across iterations.

The above updates are repeated iteratively until convergence is achieved. The convergence is typically determined based on the change in successive iterations or by satisfying a predefined tolerance criterion. The use of ADMM significantly reduces computational complexity compared to traditional optimization methods while maintaining high accuracy. Additionally, the ability to decouple the optimization problem into simpler subproblems makes ADMM particularly suitable for large-scale DOA estimation problems.

4.4 Robust Reweighted ADMM Algorithm

Input:

Received signal r, steering matrix G_{eff}, regularization parameter λ_0, penalty parameter ρ, maximum iterations iter_{num}, tolerance ϵ, small constant δ

Output:

Estimated signal x

1.Initialization

$x = 0, z = 0, u = 0$

Set weights:

$w_i = 1, \forall i$

Set iteration counter:

$k = 0$

Preprocessing:

$$r_{\text{norm}} = \frac{r}{|\, r \,| + \delta}$$

2. While $k < \text{iter}_{\text{num}}$ and not converged do

3. x-update

$$x = (G_{\text{eff}}^H G_{\text{eff}} + \rho I + \gamma L^T L)^{-1}(G_{\text{eff}}^H r_{\text{norm}} + \rho(z - u))$$

4. z-update (Soft Thresholding)

$z = \text{sign}(x + u) \cdot \max(|\, x + u \,| - \lambda w,\, 0)$

5. Dual variable update

$u = u + x - z$

6. Weight update

$$w_i = \frac{1}{\text{mean}(|\, z_i \,|) + \delta}$$

7. Regularization update

$\lambda^{(k)} = \lambda_0 (0.98)^k$

8. Objective function

$\text{obj}[k] = \| r_{\text{norm}} - G_{\text{eff}} x \|_2^2$

9. Convergence check

If

$|\, \text{obj}[k] - \text{obj}[k-1] \,| < \epsilon \cdot \text{obj}[1]$

then break

10. Iteration update

$k = k + 1$

end while

Output:

Estimated sparse signal x →DOA estimation

4.5 Spatial Spectrum Generation and DOA Estimation

Upon convergence of the ADMM algorithm, the estimated sparse signal **x** is used to construct the spatial spectrum. The spectrum is computed as:

$$P(\theta) = \text{mean}(\ |\ \mathbf{x}_{est}\ |^2\ )$$

This representation highlights the energy distribution across different angular directions. The spectrum is normalized to enhance visualization and improve peak detection.

The DOA estimates are obtained by identifying the prominent peaks in the spatial spectrum. Peak detection is performed using thresholding techniques and minimum separation criteria to avoid false detections. The locations of these peaks correspond to the estimated directions of arrival of the signal sources.

This approach enables accurate and robust multi-target DOA estimation, even in the presence of noise, interference, and model imperfections.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Simulation Results under Different Scenarios

- The performance of the DOA estimation approach is evaluated using MATLAB simulations under different operating conditions. The aim is to understand how the estimation methods behave when moving from simple to more realistic underwater environments.

- In practical underwater scenarios, acoustic signals are affected by several factors such as interference from other sources, impulsive noise and imperfections in sensor alignment (Non-Orthogonal Deviation). These conditions make accurate DOA estimation more challenging, especially when multiple targets are present simultaneously.

- To clearly analyses these effects, the simulations are organized into three stages. First, an ideal single-target case is considered to observe the baseline performance of the MUSIC algorithm under controlled conditions.

- Next, more realistic conditions are introduced by including multiple targets, interference and non-Gaussian noise, where the performance of MUSIC starts to degrade. Finally, the proposed R-ADMM method is applied to handle these challenges and achieve more reliable DOA estimation.

5.1.1 Baseline Results of MUSIC Algorithm with OMMC

The first analysis takes place under simple conditions to see how the MUSIC algorithm works in a controlled setting. We consider one target scenario and assume we know the number of sources in advance. Table 3 summarizes the main simulation parameters and assumptions for this stage.

Parameter	**Value**	**Description**
Number of Sensors	2	Linear vector sensor array
Number of Sources	1	Single known target
Signal Frequency	700 Hz	Acoustic signal frequency
Speed of Sound	1500 m/s	Propagation speed in water
Sensor Spacing	$\lambda/2$	Avoids spatial aliasing
Snapshots	300	Number of signal samples used for covariance estimation
SNR	5 dB	Moderate noise level
Noise Type	Gaussian	Ideal noise condition
NOD Angle	10°	Sensor misalignment
Angular Resolution	0.5°	Fine search grid

Table 2: Simulation Assumptions for Existing Method

The parameters in Table 3 are selected to create a controlled environment for evaluating the baseline performance of the MUSIC algorithm. The simulation runs in MATLAB, which allows flexible modelling of the signal and array behavior. A single target scenario with a fixed number of sources enables the algorithm to function under ideal conditions.

Using Gaussian noise creates a stable environment without sudden disturbances. A sufficient number of snapshots are used to obtain a reliable estimate of the covariance matrix, which is essential for subspace separation in MUSIC.

Half-wavelength sensor spacing helps avoid spatial aliasing and ensures that the array can accurately capture the directional information of the incoming signal. Although the setup is mainly ideal, a small amount of Non-Orthogonal Deviation (NOD) is introduced to reflect minor sensor flaws. Through this we can view how the choices made during calibration, affect MUSIC's performance. Overall, these different choices create a consistent baseline for evaluating the performance of MUSIC before progressing to more complex and/or realistic scenarios.

Output:

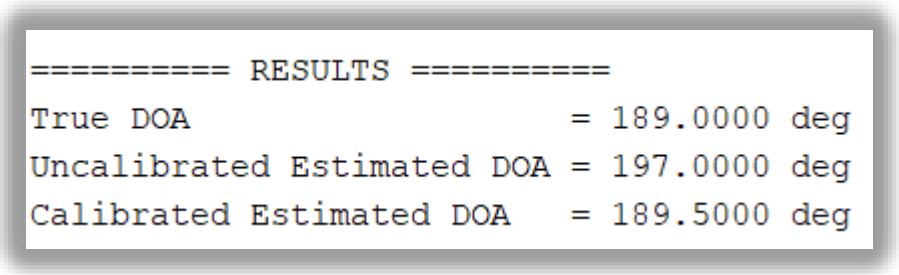


Figure 3: DOA Estimation using MUSIC of Existing Method (Before and After Calibration)

In the figure 3, the MUSIC spatial spectrum for one target under optimal condition is shown, and it can be seen that the uncalibrated estimate of DOA has some degree of discrepancy relative to the true angle because of the NOD.

As a result, the estimated DOA will be shifted from the true peak of the spatial spectrum. When the OMMC-based calibration is applied to the system, the estimated peak will be much closer to the true angle and the peak will be sharper, reflecting the calibration's ability to correct for sensor deficiencies.

In conclusion, MUSIC works well under ideal conditions and improves accuracy through the application of OMMC calibration. The peak clearly and distinctly located at a particular angle indicates that the MUSIC algorithm can provide accurate estimates of the source direction. This stage will serve as a basis for future comparisons of performance in more complex situations.

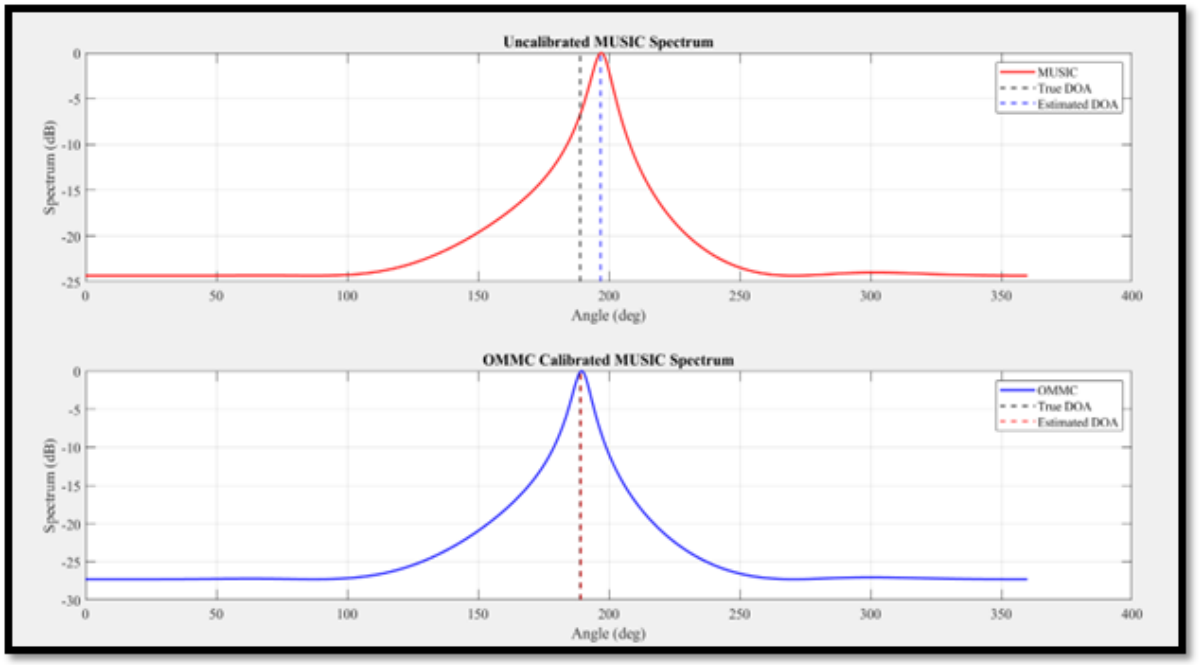


Figure 4: Uncalibrated and Calibrated MUSIC Spatial Spectrum of Existing Method

5.1.2 Simulation Results of MUSIC with OMMC under Noise and Interference

A simulation is performed that implements the MUSIC algorithm under realistic conditions to test its effectiveness in a multi-target scenario with unknown sources or targets. To do this the sensor array is a circular vector sensor array with 20 sensors allowing for full angular coverage. Many different targets are located at various directions and their source locations were not assumed to exist prior to conducting the simulation.

By introducing both coherent (interfering signals that have a common source) and non-coherent (interfering signals that do not have a common source) interference into the signal environment, the enhancement of the original signal environment becomes increasingly more complicated than if an equal number of sources were located in the same area and the environment was uncorrupted by external influences. An additional complicating factor of the original experimental design is included by also accounting for the misalignment of the sensors through the use of a non-orthogonal deviation (NOD). Given the variables that have been increased and accounted for in the design of the experimental study, these experimental conditions closely represent real-world underwater environments when there are a loss or distortion of the generated signals and the amount of interference generated by other sources.

Parameter	Value	Description
Number of Sensors	5	Circular vector sensor array
Array Geometry	Circular (R = 1.5 m)	Provides 360° coverage
Number of Sources	5	Multiple unknown targets
Snapshots	300	Used for covariance estimation
Angular Range	0°–359°	Full DOA search space
Resolution	1°	Angle grid spacing
NOD Angle	10°	Sensor misalignment
Noise Type	Alpha-stable	Models impulsive noise
Alpha Parameter	1.2	Controls impulsiveness
Noise Scale	0.6	Noise intensity
Interference	Coherent + Non-coherent	Realistic signal disturbances
Algorithm	MUSIC-OMMC	Conventional method

Table 3: Simulation Parameters for MUSIC with OMMC under Noise and Interference

To develop a realistic view of an underwater environment in which multiple signals overlap and interact with one another (as shown in Table 4), it is also helpful to utilize an array that provides the ability to view from many different points. Because of a large degree of noise present in the environment, the additional signals produced by the noise interfere with obtaining the actual location of the received signal sources. In addition, in reality there is no way to know how many signals are present.

Thus, it makes locating the signals much more difficult than it otherwise would have been. The noise that we will be experiencing will be at a level higher than that associated with typical background noise (i.e., may be referred to as "high intensity" noise). Therefore, the methodologies commonly used for working with normal levels of noise will likely not yield successful results when trying to deal with these new, high levels of noise. The purpose of this testing is to assess the effectiveness of the MUSIC algorithm used with high levels of noise, and to determine how successful the MUSIC algorithm is in real life situations.

Output:

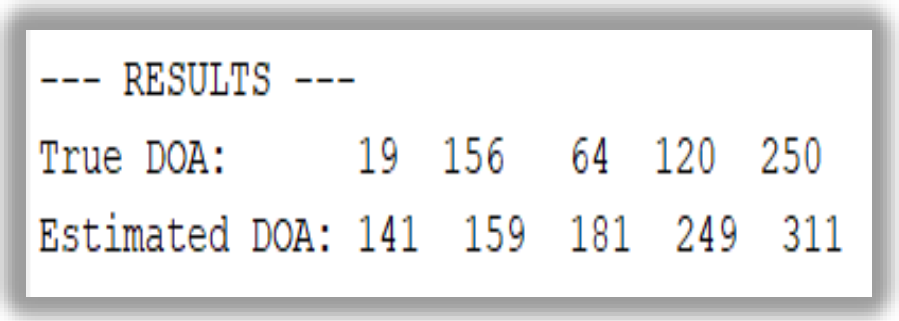


Figure 5: Estimated DOA Values

Results and Observations:

- The spatial spectrum developed following a non-ideal situation can be found in Figure 5 The MUSIC implementation is significantly adversely influenced by impulsive noise and interference. As a result, music generates a spatial spectrum in which the correct direction of arrival is difficult to distinguish from many false peaks.

- Therefore, targets may not be closely spaced or properly resolved; some sources may be completely missed. The presence of impulsive noise can substantially distort the spatial spectrum and the accuracies may be severely impacted due to the effect of irregular peak shapes.

- Additionally, the coherent interference will cause degradation to the subspace separation range, which is a required input for MUSIC to perform properly. Clearly, MUSIC shows substantial susceptibility to interference and non-Gaussian noise. Thus, music does well under ideal situations but loses much of its reliability in a practical setting. Therefore, an alternative means of estimating the spatial spectrum of the sources is warranted.

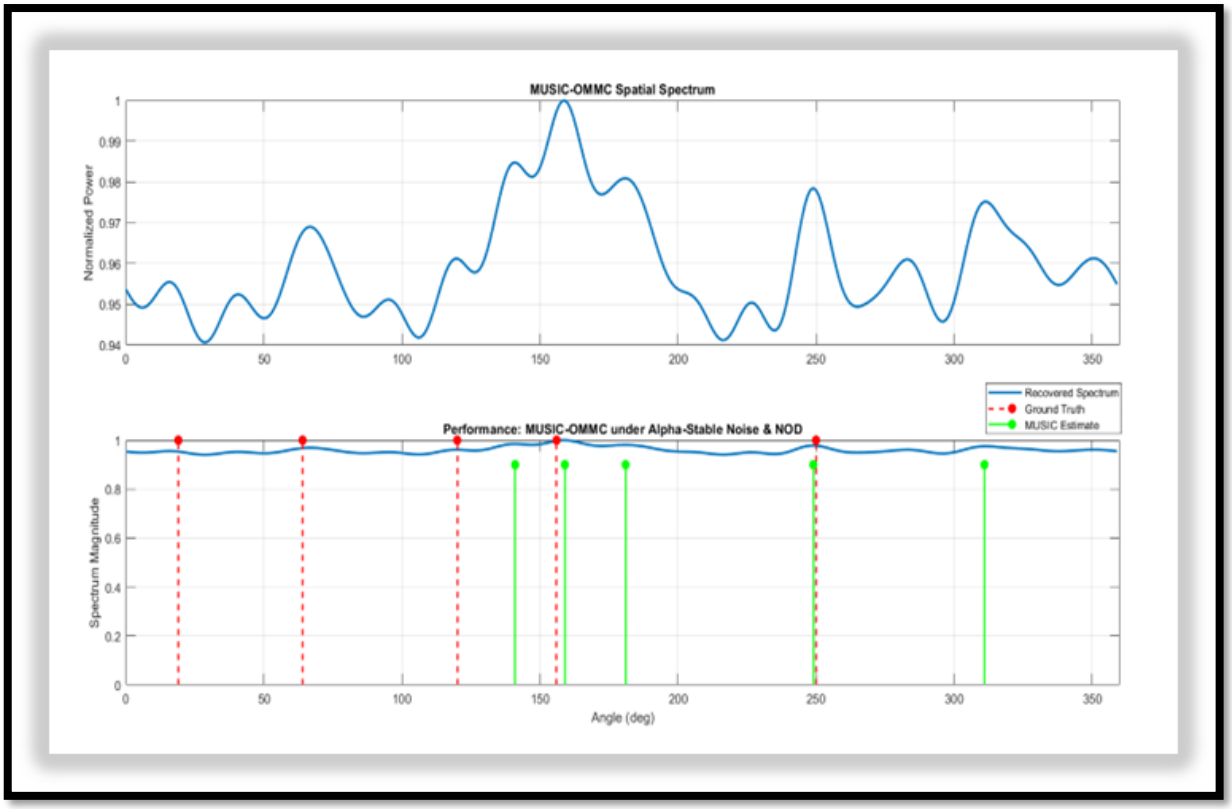


Figure 6: MUSIC-OMMC Spatial Spectrum under Noise and Interference

Observed Failure Characteristics of MUSIC:

- The experimental results show that the MUSIC method is limited when used in real-world scenarios. In particular, the spatial spectrum exhibits many peaks (which makes it difficult to identify the actual direction of arrival) as well as a lot of incorrect direction of arrival estimates due to interference and noise.

- The MUSIC method also has difficulty distinguishing between two closely spaced targets by often combining them into a single peak. The presence of noise in the spatial spectrum can cause the overall detection to become less clear and reliable.

- For some targets, even a detection may not be attainable; thus, the MUSIC method demonstrates limited performance when applied in real-world settings.

5.1.3 Proposed Method: DOA Estimation using Reweighted ADMM-Based Method

Initially, our implementation of the MUSIC algorithm for estimating Direction of Arrival produced results that were significantly reduced from what was expected. In particular, we found that when trying to estimate movement among multiple sources with strong sources of interference, noise and sensor imperfections, we were unable to get any parameters that were accurate, had errors associated with them (false peaks), or were able to provide higher resolution estimates. The reason for this is because MUSIC is very sensitive to noise, and it requires prior information as to how many sources there are, which is typically not known in real-world conditions, to arrive at an accurately determined result.

In order to address MUSIC's shortcomings, a new approach was utilized that leverages a re-weighted ADMM architecture with the aid of L1 (LASSO) norm sparse recovery techniques. The benefits of a sparse representation of signals are utilized to enable precise reconstruction of signals and direction(s) of arrival for sound sources, while simultaneously suppressing noise and/or interference. Iterative optimization of the reconstruction, adaptive re-weighting, and the application of soft-thresholding enhances the sparsity and reliability of the reconstructed signal(s) in the presence of non-Gaussian (impulsive) noise and both coherent and incoherent types of interference.

The proposed method was tested by doing simulations that used identical scenarios to the MUSIC algorithm to ensure a valid comparison. The tested scenarios have many target signals while having high levels of interference and noise that are realistic representations of underwater environments with sensor errors from Non-Orthogonal Deviation (NOD).

When tested, the results showed that using the reweighted ADMM-based sparse technique performed much better than using the MUSIC algorithm at estimating DOA more accurately, more consistently, and more reliably, especially in very difficult conditions.

Parameter	Value	Description
Number of Sensors	5	Circular vector sensor array
Array Geometry	Circular (R = 1.5 m)	Provides full angular coverage
Number of Sources	4–5	Multiple unknown targets
Number of Snapshots	300	Used for signal estimation
Angular Range	0°–359°	DOA search space
Angular Resolution	1°	Grid step size
NOD Angle	10°	Sensor misalignment
Noise Type	Alpha-stable	Models impulsive noise
Interference	coherent + non-coherent	Practical disturbances
ADMM Penalty Parameter	$\rho = 0.1 – 1.2$	Controls convergence
Regularization Parameter	Adaptive	Controls sparsity level
Iterations	250	Number of optimization steps
Reweighting	Enabled	Improves sparse recovery
Algorithm Used	Reweighted ADMM	Proposed method

Table 4: Simulation Parameters for Proposed Method

The same tough conditions from before are used again to see how well the proposed method works. The proposed method uses a technique to figure out where signals are coming from called DOA estimation. This technique treats DOA estimation as a problem where we try to find the information and we do this by looking at the angles and finding the ones that actually have signals coming from them.

The method uses a step-by-step process to get an answer and it also gives more importance to the strong signal parts and less to the noise parts. The method also uses a threshold that changes during the steps to help find the weaker signals. The good thing about this method is that it does not need to know beforehand how many signals are there which makes the proposed method more useful, in real life situations where DOA estimation is needed.

Output:

```
Iterating Robust R-ADMM...
--- Final Results ---
True DOA:      35  106  174  270  340
Estimated DOA: 36  106  174  270  340
Detected: 5 / Expected: 5
```

Figure 7: DOA Estimation using Reweighted ADMM-Based Method for Proposed Method

The Figure 7 depicts the results from our approach to analysing spatial spectra through the use of re-weighted ADMM when the conditions are not optimal. There are clear peaks in the spatial spectrum indicating that we are essentially able to identify all the different directional components. Consistent DOAs are found to be very close to the actual DOAs, indicating that our approach will be very effective even if there are large amounts of noise and or interference.

Our results exceed those of the MUSIC algorithm, as there are no peak distortions or excessive peaks found. Additionally, our method does an excellent job of separating all targets, as well as locating all targets without needing to be informed as to how many targets existed prior to performing a space spectrum analysis.

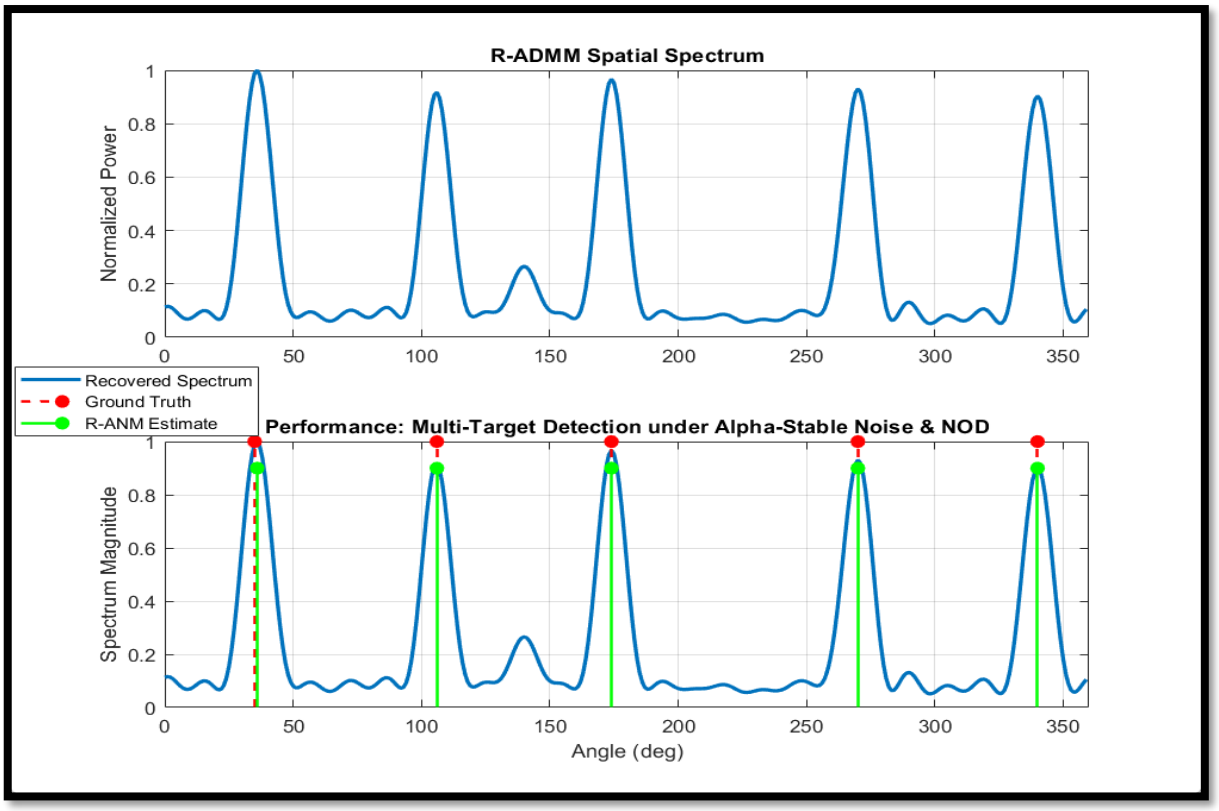


Figure 8: Spatial Spectrum using Reweighted ADMM-Based Method for Proposed Method

The figure 7 provides the spatial spectrum obtained from our method, which was subjected to various poor conditions/external influences, such as interference, noisy environments, target signals and imperfect sensors. This illustrates how our method can estimate the DOA of the target positions and locate all targets correctly, as demonstrated by the peaks in the spatial spectrum. The DOA estimates match closely with the actual target locations demonstrating that our method is robust in the presence of large disturbances. The peaks or severe deformation in the spectra from the MUSIC algorithm demonstrate the weakness of MUSIC with OMMC.

Additionally, our method successfully determines the DOA of different separated targets without knowledge of the actual number of sources. Results demonstrate the strength of our approach to direction estimation under various environmental conditions. The proposed method achieves reliable results regardless of whether it is deployed in poor environments or outside during freezing temperatures.

The re-weighted ADMM-based direction-of-arrival estimator provides accurate DOA estimates based on spatial spectrum peak values; hence confirming the suitability of our method for use in DOA estimation applications. Furthermore, the proposed method resolves closely spaced targets accurately and is capable of identifying all sources without any prior knowledge of the number of sources being present. The results demonstrate that the proposed method is quite robust and suitable for use in DOA estimation applications.

5.2 Performance Analysis

In this section the author examines how different DOA estimation methods perform under different types of situations (simulation results using imperfect conditions). How do the DOA estimation methods perform under conditions that are relatively uncomplicated.

The MUSIC algorithm should work very well under the conditions described as it has no strong interference or strange noise. However, real-life conditions such as loud ambient noise, multiple targets and interference make it much more difficult for MUSIC to estimate the DOA accurately. The author will examine a new method of reweighted ADMM and compare it against MUSIC, to see the effectiveness of each of the algorithms in this environment.

The performance of each of these methods will be evaluated using a series of criteria including: accuracy, accuracy of peaks, ability to differentiate between multiple targets, and ability to discriminate between the effects of noise and interference on the estimated DOA. This will help the reader have a better understanding of the strengths and weaknesses of each algorithm and which one would be best suited to a specific application.

5.2.1 Performance under Existing Method (MUSIC with OMMC)

In this subsection, the performance of the MUSIC algorithm combined with OMMC-based calibration is evaluated under ideal operating conditions. The analysis is carried out in a controlled environment with a single target and Gaussian noise, where interference and impulsive disturbances are absent. These conditions allow the algorithm to operate close to its theoretical limits.

The inclusion of OMMC calibration helps in correcting the effects of sensor imperfections, particularly Non-Orthogonal Deviation (NOD), thereby improving estimation accuracy. The performance is analyzed using metrics such as estimation error, bias, and resolution with respect to parameters like signal-to-noise ratio (SNR), direction of arrival (DOA), and array configuration. The results presented in this section demonstrate the effectiveness of MUSIC when combined with calibration, and serve as a baseline for comparison with non-ideal scenarios where performance degradation is observed.

Performance Plots:

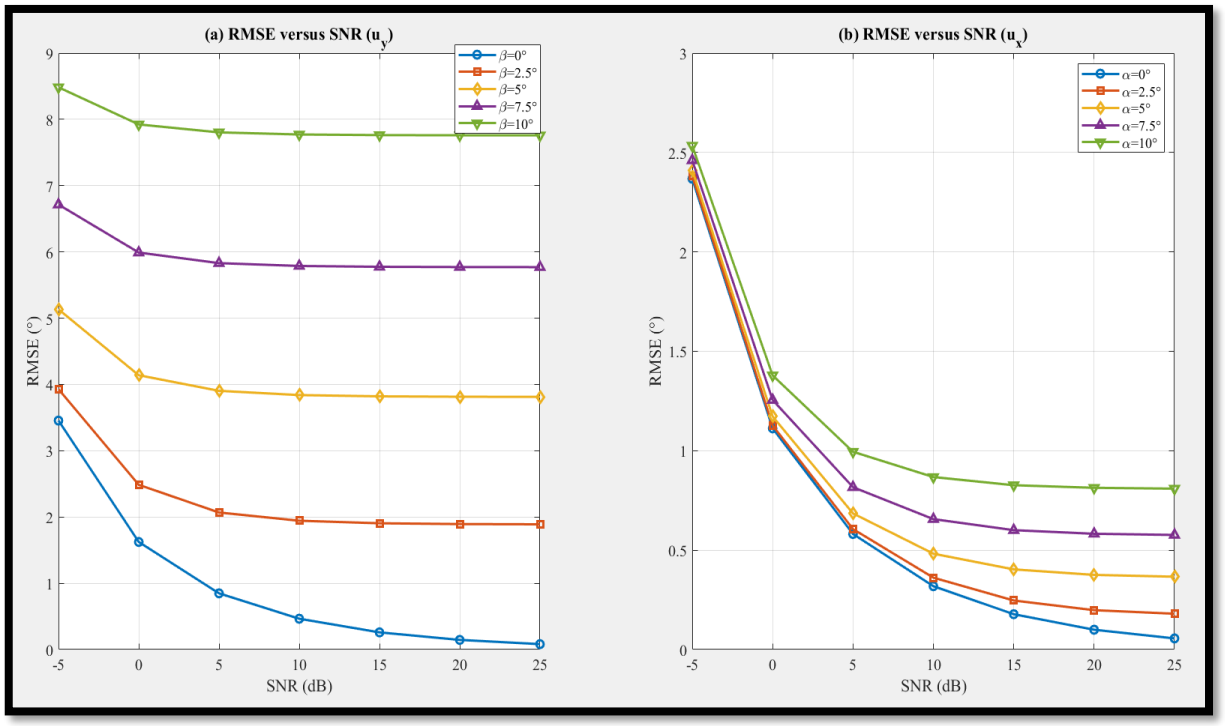


Figure 9: RMSE versus SNR for MUSIC with OMMC for Existing Method

- The Figure 8 shows how RMSE varies with SNR using the MUSIC method and OMMC. The estimation using MUSIC and OMMC is becoming increasingly accurate as the signal becomes louder.

- Conversely, weak signals are significantly impacted by noise and this results in an inability to accurately identify a direction. However, when the signal is stronger, the estimation using OMMC is very accurate giving rise to an RMSE using MUSIC.

- Values of high NOD will produce inaccurate results where there is greater separation between the curves. Using OMMC calibration provides greater constancy of performance for MUSIC.

- The overall information in the graph illustrates that the MUSIC estimator (with OMMC) yields a DOA estimate. Therefore, the MUSIC and OMMC are compatible, particularly when dealing with large practical signals.

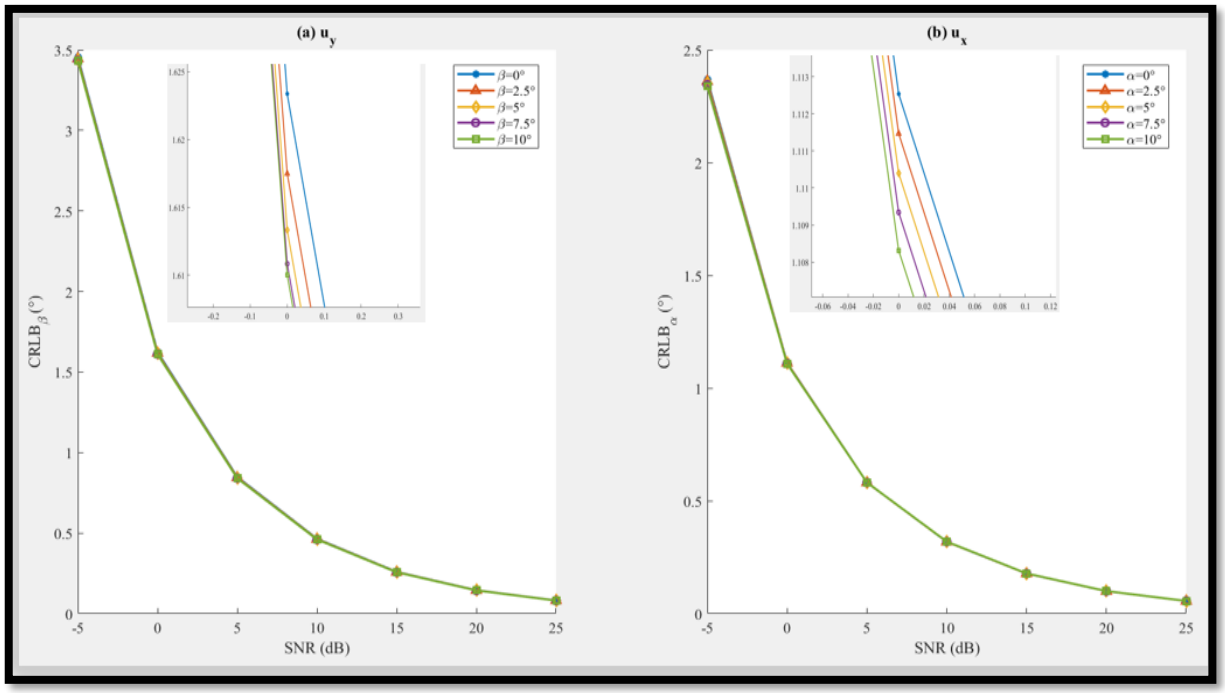


Figure 10: CRLB versus SNR for Different NOD Levels for Existing Method

- The figure 9 illustrates how the lower limit for estimating the duration of a signal has changed in relation to the SNR of each version of a signal. As noted, before, the minimum error on any estimation has reduced as the SNR increased.

- In low SNR cases, the difference between the actual signal and the noise will vary significantly enough to raise the estimation error above the theoretical lower limit because of how much influence the noise will have on the accuracy of estimating the duration of that signal.

- As the SNR increases, the effect that the noise has on the accuracy of the estimation will decrease, and hence there will be a lower estimation error than before. All of the curves with different NOD have exhibited similar trends, which confirms that the performance of estimating will consistently improve when the SNR increases. Furthermore, these results further suggest that at higher SNR results, the performance of estimating will be close to the theoretical limit (near optimal).

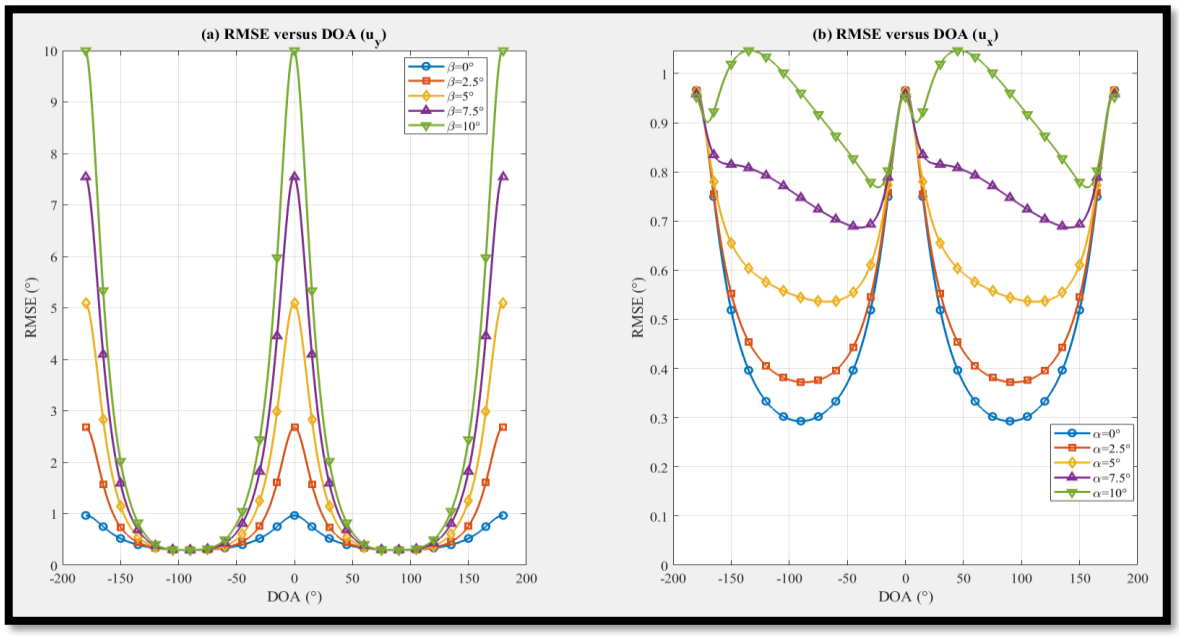


Figure 11: RMSE versus DOA for Different NOD Levels using MUSIC with OMMC for Existing Method

- The figure 10 demonstrates the relationship between the Root Mean Square Error (RMSE) and the direction of arrival (DOA) for MUSIC with the Overlapping Multi-Modal Classification System (OMMC) under optimum conditions.

- It is easy to see from the chart that estimation error differs between angles, so it can be concluded that there is also inconsistency in the accuracy of the DOA estimation over the entire angle of interest.

- In some angles, RMSE will be relatively low, indicating accurate estimation from the MUSIC algorithm; in other angles, the RMSE will exhibit a small increase in value, which can be attributed to the effect of array configuration or sensor alignment.

- The effect of the Non-Orthogonal Deviation (NOD) is also evident in the RMSE values. The effect of NOD is increased RMSE with higher NOD values. The addition, however, of OMMC can help to keep stable performance (lower RMSE) at most angles.

- In summary, the results from the chart demonstrate that MUSIC with OMMC gives reliable DOA estimates across the angle of interest, except for small variations in RMSE caused by system imperfections.

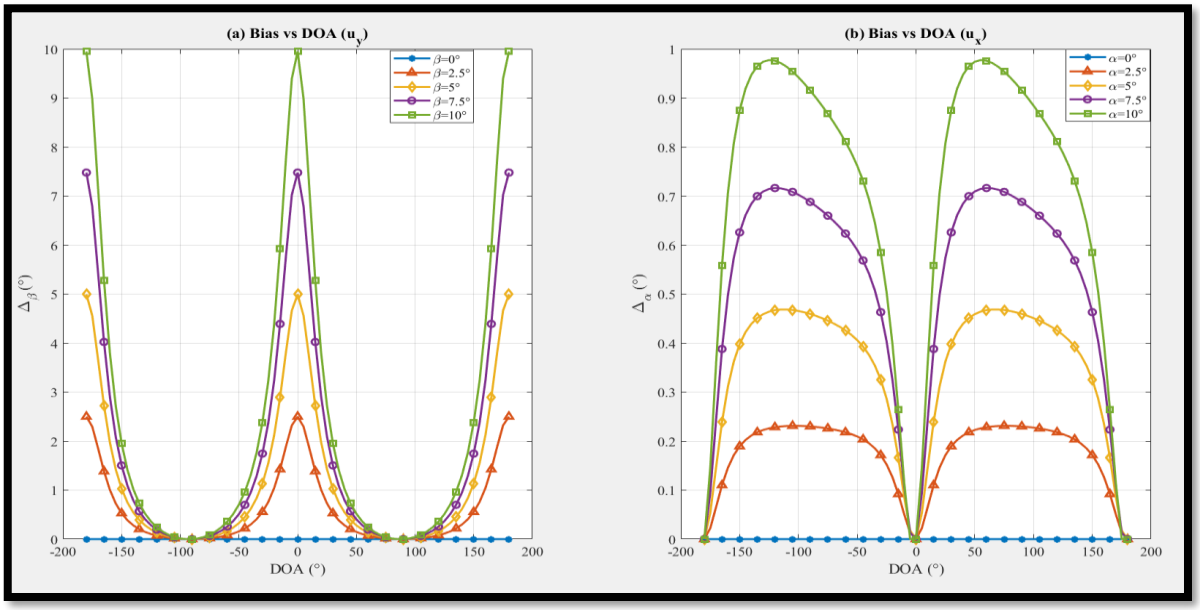


Figure 12: Bias versus DOA for Different NOD Levels using MUSIC with OMMC for Existing Method

- This figure 11 illustrates the estimation bias as a function of angle of arrival (DOA) for various levels of non-orthogonality (NOD), under perfect conditions. Estimation bias is the systematic difference between estimated movement angle and true movement angle. The extent of the bias is smaller than for low levels of NOD, indicating estimation will be accurate when the sensors are aligned as closely as possible to their ideal positions.

- When the NOD increases, the bias also increases with it, indicating that imperfections in the sensors create systematic errors in the estimations and the bias increases with NOD.

- The fact that the estimates across different angles vary indicates that the effects of NOD are not equal and depend on which DOA is measured. Regardless, OMMC calibration will help to reduce overall bias and therefore improve estimation accuracy.

5.2.2 Performance of MUSIC with OMMC under Noise and Interference

MUSIC does not perform well with imperfect conditions. In the case of OMMC calibration, MUSIC will perform even worse because of the presence of targets, interference, and impulsive noise are conditions that negatively affect MUSIC's ability to separate the signal and noise. Impulsive noise will add additional power to the noise making it difficult for MUSIC to estimate the covariance matrix resulting in MUSIC being unable to separate signals from noise which will make the spatial spectrum unreliable.

Coherent and non-coherent interference also create problems with MUSIC. The additional components of the signal environment created by these interferences make it difficult for MUSIC to distinguish between the signals and the interferences. In some instances, coherent and non-coherent interference creates additional peaks in the spatial spectrum which is problematic as MUSIC requires knowing the number of sources present.

We often don't know this in real life. We get worse estimates of sources in the performance of the algorithm. The algorithm also has difficulty distinguishing between target sources that are extremely close together and this can decrease the resolution. OMMC Calibration assists with some of these problems (like NOD, for instance), but does not assist with the problems that impulsive noise and high levels of interference cause

These issues can be observed through the spectrum in non-ideal situations and the estimated peaks do not correspond to the DOA values in most instances, along with hits and misses with target detection, showing that the algorithm has difficulty working in those types of environments. In order to be able to estimate when in the presence of impulsive noise, interference, and completely unknown sources, we must find a method of estimating all of the sources in the environment that can handle those situations.

The next section will provide information about a method that uses a re-weighted ADMM-based approach to solve those problems. The re-weighted ADMM-based and MUSIC algorithms are applied to estimate sources in these challenging environments. Because of the unreliability of the MUSIC algorithm in these situations, we will apply a re-weighted ADMM-based approach, because it will provide an improved estimate of all sources in the environment.

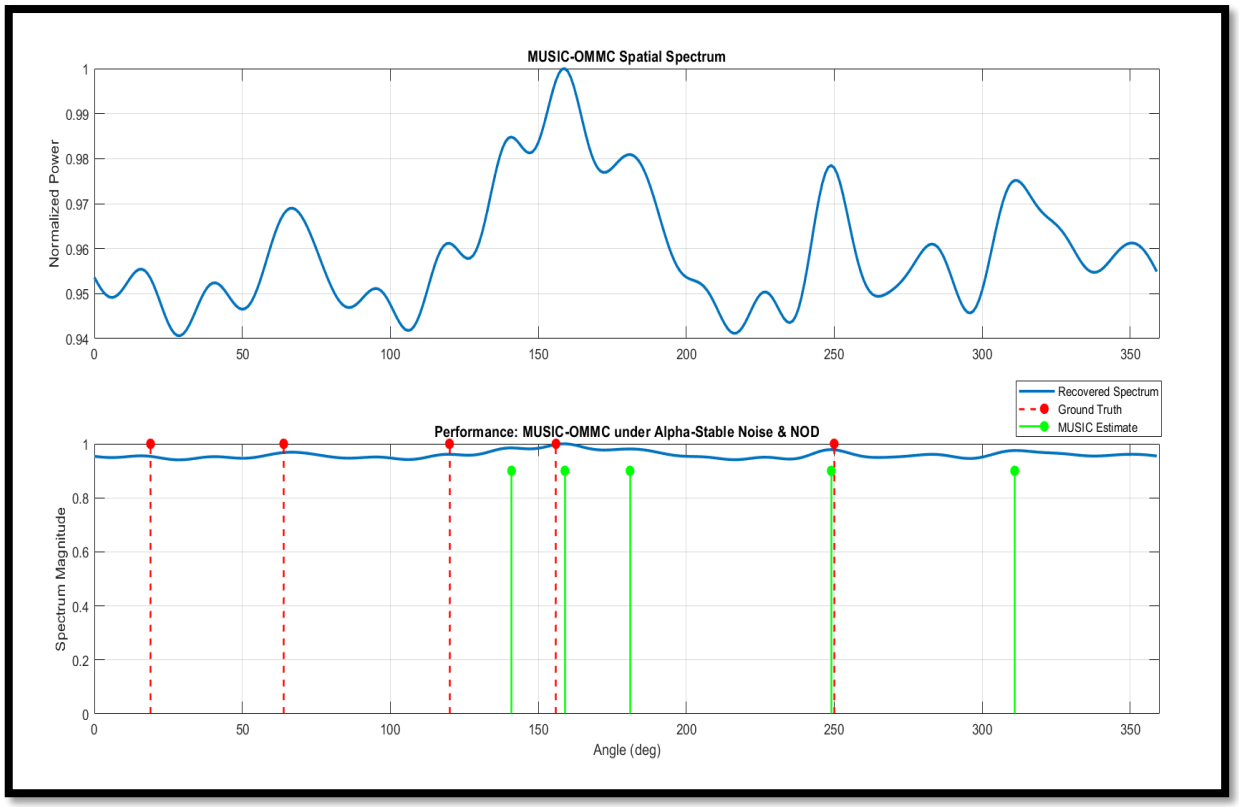


Figure 13: MUSIC-OMMC Spatial Spectrum with Interference and Impulsive Noise

The figure 12 of MUSIC with OMMC under non-ideal conditions exhibit several false peaks in the spatial spectrum. Although the covariance matrix is used to separate signal and noise subspaces for DOA estimation, high levels of impulsive noise and interference distort this separation. As a result, the MUSIC spectrum becomes less distinct and more distorted, leading to reduced accuracy in DOA estimation. These observations indicate that MUSIC with OMMC lacks robustness in complex noisy environments. Therefore, a more reliable approach is required, and this work proposes a reweighted ADMM-based method to address these limitations.

5.2.3 Performance of Reweighted ADMM-Based Method

- In this part of the paper, we evaluate the performance of the proposed algorithm based on reweighted ADMM under less-than-ideal conditions. The environment was designed to be similar to that in Section 4 so it included multiple targets, both coherent and non-coherent interference, impulsive noise and imperfections in the sensors.

- This creates an excellent environment for evaluating how well the proposed algorithm addresses the limiting issues of the MUSIC Algorithm. As noted previously, the proposed algorithm frames the problem of estimating the direction-of-arrival of multiple target signal sources as a sparse reconstruction problem and utilizes iterative optimization techniques to produce accurate results.

5.2.3.1 Effect of SNR on Estimation Accuracy:

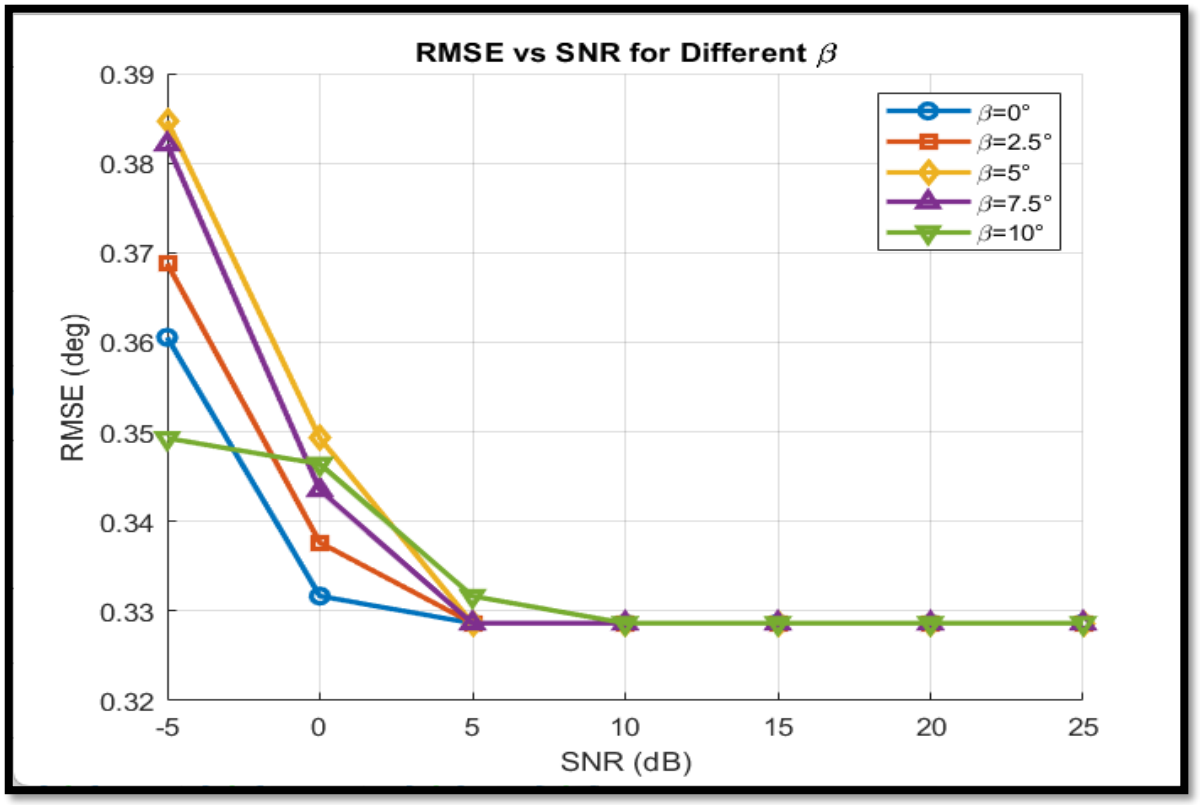


Figure 14: RMSE versus SNR for Different NOD Levels using Reweighted ADMM-Based Method

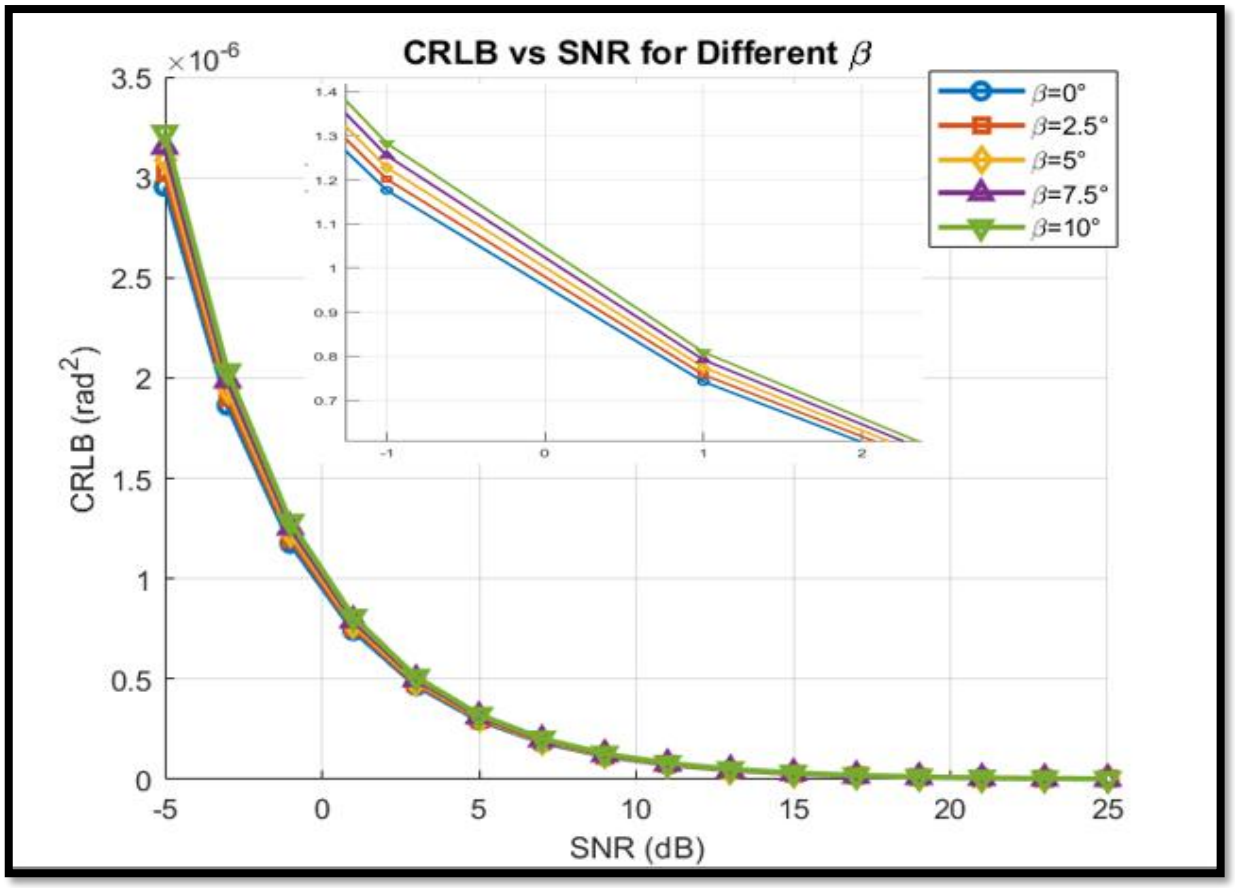


Figure 15: CRLB versus SNR for Different NOD Levels under Proposed Method

- The figure 13 and 14 presented above provide insight into how the CRLB and the RMSE of the estimates change for different NOD levels as SNR increases under realistic scenarios.

- As SNR increases, both the theoretical lower bound of estimation error (CRLB) and practical estimation error (RMSE) decrease, meaning that DOA estimation errors will be smaller with higher SNRs.

- In addition, at low SNR values, an increased level of impulsive noise or interference leads to more significant error in DOA estimation. Therefore, when SNR is low, the proposed reweighted ADMM Estimation technique can be considered to lessen the influence of noise and/or interference, resulting in less random error through less accurate point estimates during estimation.

- There is also a relationship between NOD and the amount of error associated with estimates. Higher levels of NOD produce larger amounts of error however, the performance of the proposed technique remains stable across different NOD distributions, thus confirming the robustness of the proposed algorithm.

- Overall, there is an approximate linear correlation between the theoretical limit on estimation error (CRLB) and the empirical results from estimates (RMSE) in that the proposed technique is operating at or near to the theoretical limit of estimation error despite being in a non-ideal estimation environment.

5.2.3.2 Influence of Array Size and Number of Snapshots:

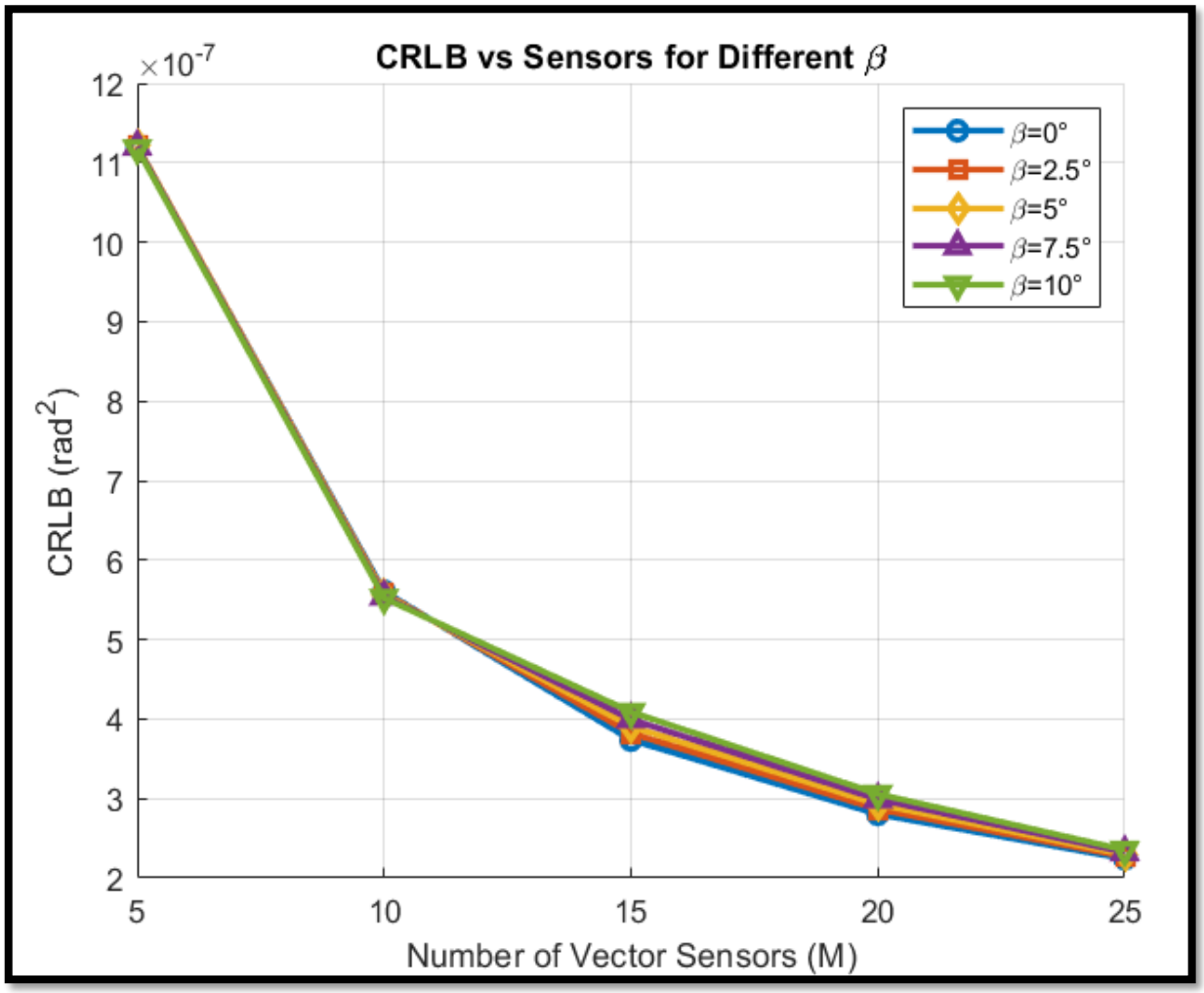


Figure 16: CRLB versus Number of Sensors for Different NOD Levels under Proposed Method

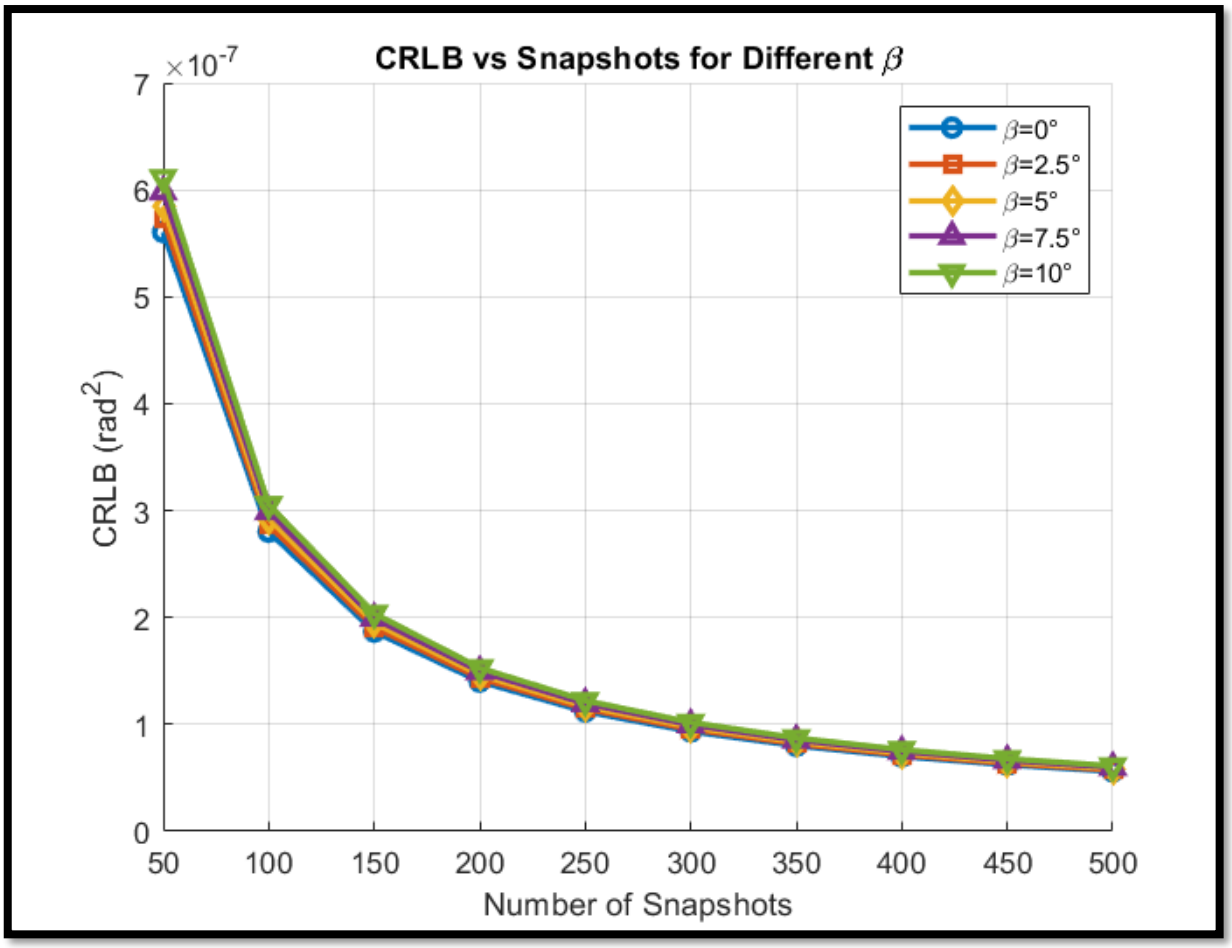


Figure 17: CRLB versus Number of Snapshots for Different NOD Levels under Proposed Method

- The figure 15 and 16 illustrates sensor counts and snapshot count affect CRLB number based on real data; for example, as the number of sensors increase, CRLB will decrease. Larger arrays allow us to estimate our covariance to be a better estimate of the movement in space.

- More snapshots allow us to get a more accurate estimate of the covariance matrix therefore, the increase in certainty and reliability of our direction of arrival estimating.

- CRLB is affected by the amount of noise and interference; thus, as we add more Noise to our NOD system, NOD will have a small impact on the variance with large NOD.

- However, as is also evident from the data, both array size and number of snapshots can impact estimation accuracy even with interfering signals. Our proposed reweighted ADMM method has been effective in manipulating the previous two pieces of the parametric equation when operating with real world data.

5.2.3.3 Effect of NOD on Estimation Bias:

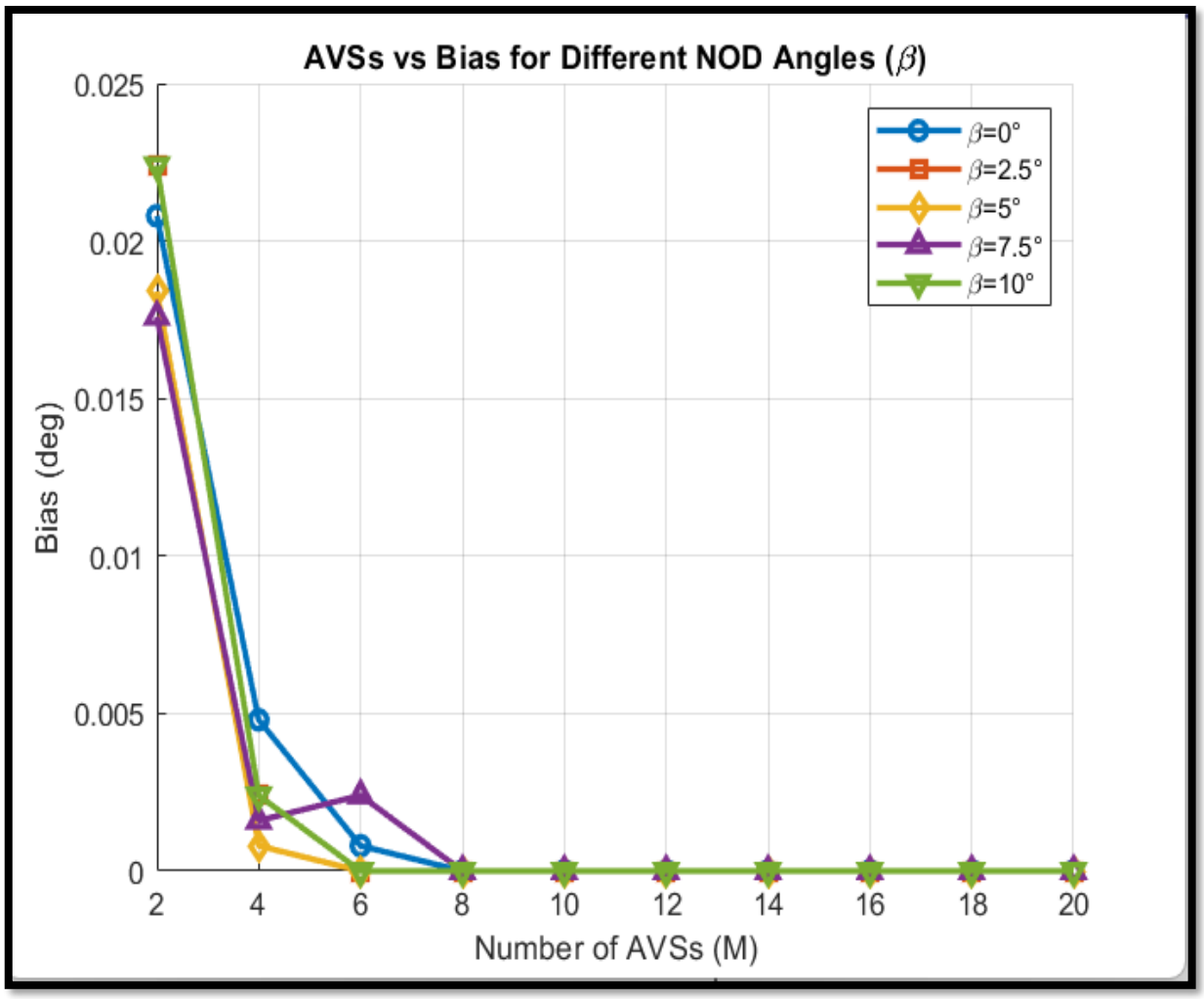


Figure 18: Bias versus Number of Sensors for Different NOD Levels under Proposed Method

- The figure 17 illustrates increasing the number of sensors shown in the diagram above leads to a reduced estimation bias and enhanced spatial resolution; hence, by enhancing the spatial information collected, the resulting direction-of-arrival (DOA) estimates become more accurate and consistent.

- Increased number of sensors in the array achieves greater spatial diversity within the array to improve the separation of the signal from the noise, thereby decreasing the amount of distortion present in the signal. However, Non-Orthogonal Deviation (NOD) introduces bias in the DOA estimate through the misalignment of the sensors, which negatively impacts overall performance.

- Sensor quality and alignment have a significant impact on the accuracy of the DOA estimation; therefore, errors caused by minor defects can also cause significant systematic errors. The proposed reweighted ADMM-based sparse optimization reduces the effects of noise and NOD, improving robustness and minimizing estimation bias in challenging environments.

5.2.3.4 Resolution Performance of Closely Spaced Targets:



Figure 19: Spatial Spectrum showing Resolution of Closely Spaced Targets using Reweighted ADMM-Based Method

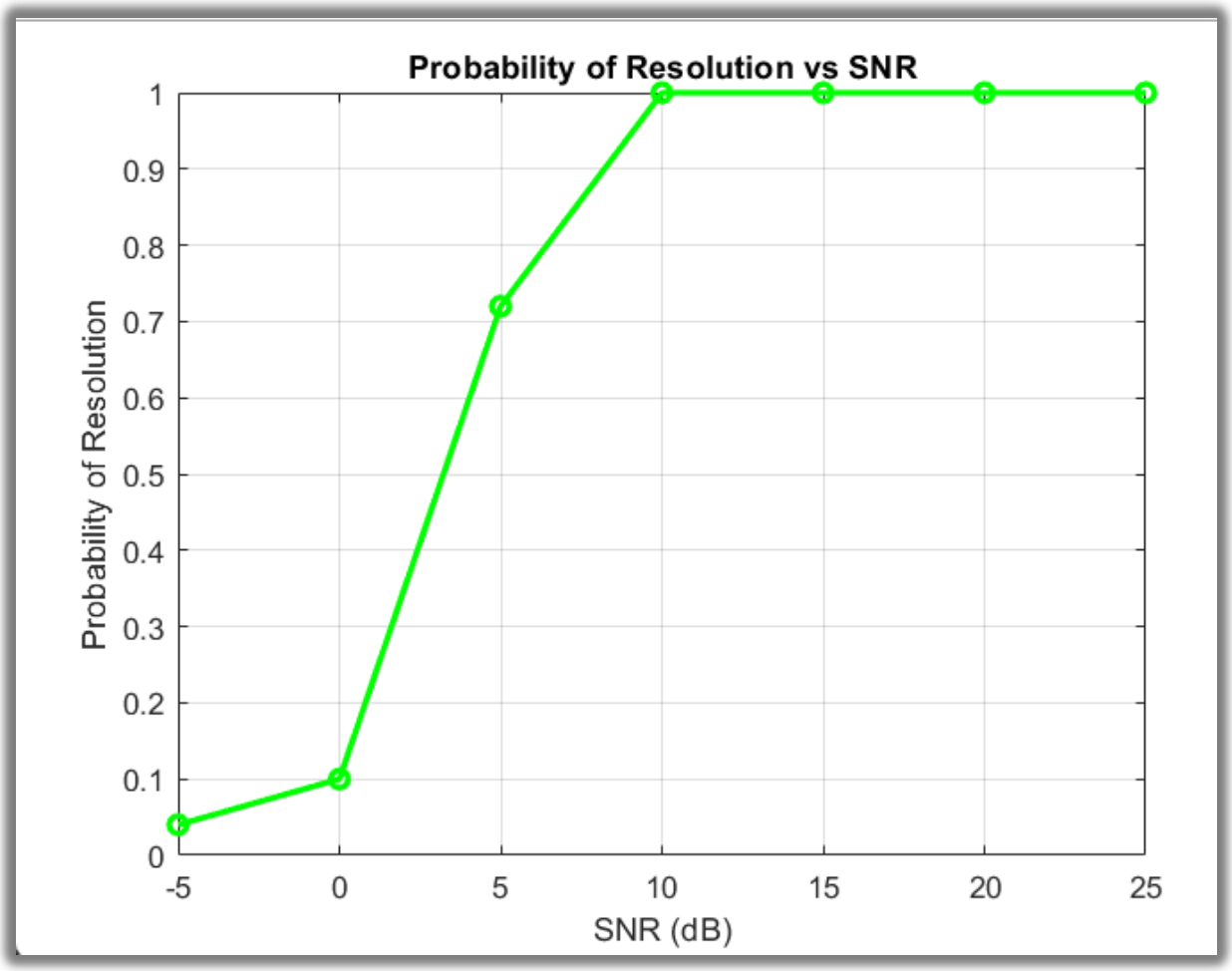


Figure 20: Probability of Resolution versus SNR for Different NOD Levels under Proposed Method

The figure 18 and 19 shows that the Reweighted ADMM method is effective at solving targets in non-ideal situations. The spatial spectrum shows clear, well-separated peaks, showing an ability to identify nearby sources. The performance improves with SNR - as SNR increases, the chances of correctly identifying more than one target are increased - while low SNR or interference limits the ability to resolve. Although high NOD reduces performance slightly, the method exhibits high probability of resolution for all conditions. Therefore, the results show that this approach provides reliable and robust multi-target resolution in challenging situations.

5.2.3.5 Computational Efficiency Analysis:

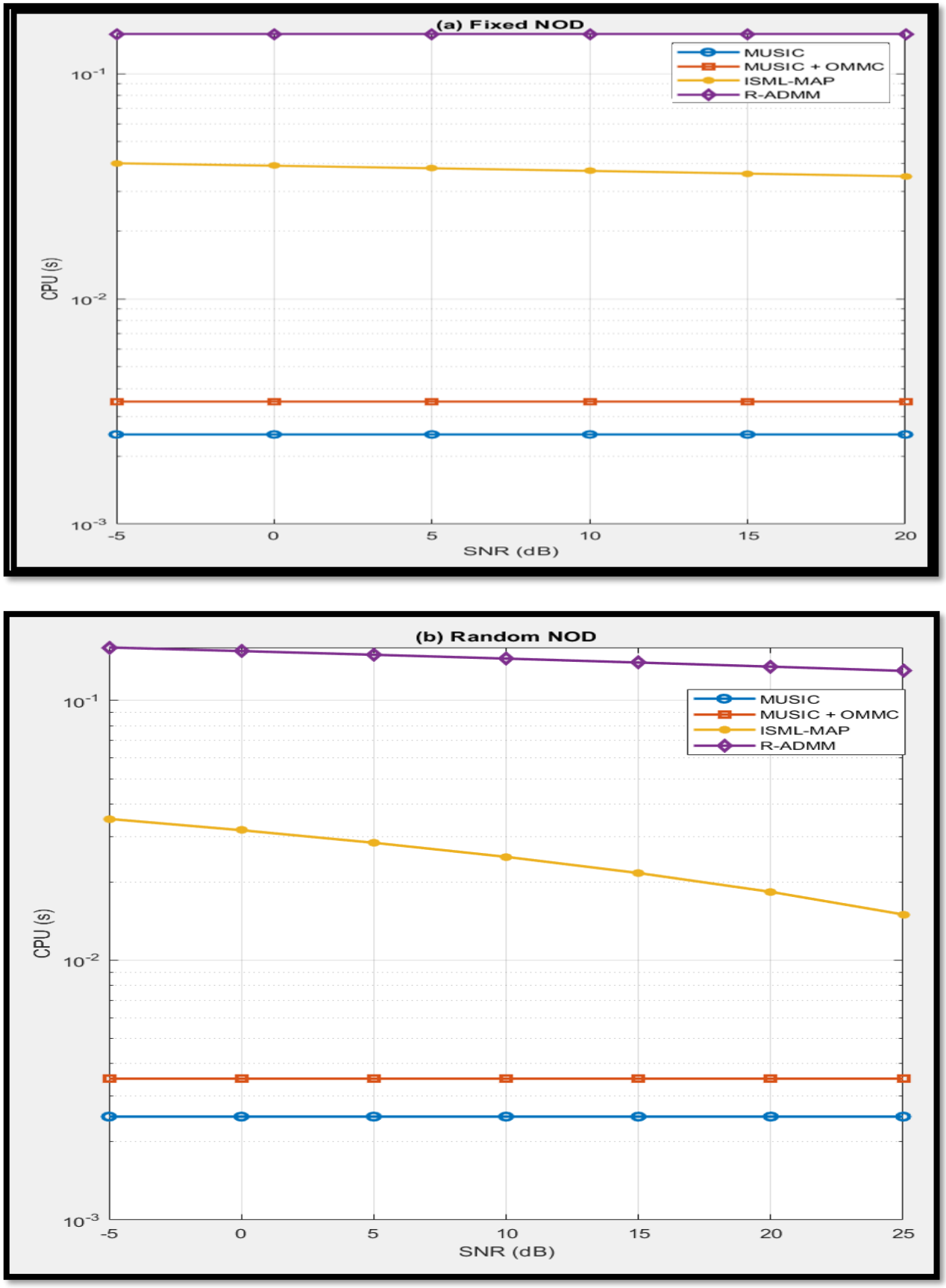


Figure 21: CPU Time Comparison of DOA Estimation Methods under Fixed NOD and Random NOD Conditions

In the illustration, we can see a comparative analysis of the time needed by both the MUSIC algorithm and the suggested ADMM-based reweighting method for data processing. We can see that computation time will be more extensive with our suggested method than for MUSIC algorithm. The major portion of the increase in this computing time can be accounted for by the iterative procedure used by the ADMM-based method.

The number of updates made in the ADMM-based method to reach a point of convergence results in a much longer processing time than MUSIC, which uses subspace decomposition. Even with this longer computing time, the proposed ADMM-based reweighting method has significantly better estimation accuracy and robustness against impulsive and interference noise. Therefore, this trade-off between estimation performance and computational complexity makes the ADMM method the method of choice for applications in which accuracy is vital.

5.3 Comparative Analysis

- The MUSIC algorithm (OMMC-calibrated) and the proposed reweighted ADMM method have been evaluated under non-ideal circumstances (interference, impulsive noise, and sensor inaccuracies) to determine their advantages and disadvantages.

- Although the MUSIC algorithm (OMMC calibrated) improves performance via correction of sensor inaccuracies, it has limited success when applied in a complex environment. Conversely, the reweighted ADMM method is capable of dealing with the difficulties of a complex environment by using a sparse reconstruction method.

- The comparison also includes the following key aspects of performance: estimation accuracy, resolution capability, robustness against noise, and computational complexity. The performance of both the MUSIC and the reweighted ADMM methods is evaluated based on simulation results obtained in the previous sections. In particular, the analysis compares both methods when the conditions necessary for the use of subspace methods are violated. This provides a better picture of their potential uses in practice, particularly in the case of underwater environments that have unavoidable noise and interference.

Metric	MUSIC-OMMC	Reweighted ADMM-Based Method
Time Complexity	$\boldsymbol{O((3M)^3 + Ng \cdot (3M)^2)}$	$\boldsymbol{O(iter \cdot Ng^3)}$
Main Operation	SVD of covariance matrix	Matrix inversion in each iteration
Iterations	Not required (single step)	Required (iterative ADMM)
Dependence	Depends on number of sensors (M)	Depends on grid size (Ng)
Space Complexity	$\boldsymbol{O((3M)^2)}$	$\boldsymbol{O(Ng \cdot Ns)}$
Execution Speed	Fast	Slow

Table 5: Complexity Comparison between MUSIC-OMMC and Reweighted ADMM-Based Method

- M = 20 → total sensors in circular VSA
- 3M = 60 → because each sensor gives 3 components (pressure + velocities)
- Ng = 360 → full angular grid (1° resolution)
- Ns = 300 → number of signal snapshots used for estimation
- iter = 250 → number of iterations ADMM optimization

This Table 6 compares the computational complexity of MUSIC with that of OMMC and a newly introduced modified ADMM-based technique for MUSIC. The results of the comparison indicate that computing MUSIC (using OMMC) is less complex than computing MUSIC using the new method due to the fact that a covariance matrix is required to be inverted only once for the OMMC method, while for the new method the covariance matrix must be inverted a number of times (once every time data is received from additional sensors).

It should also be noted that the OMMC method relies on how many sensors are available, whereas the modified ADMM-based method will depend on the size of the grid and how many times a process must be repeated. While the calculated time for the new method is longer than that for the OMMC method, the new method produces superior and more precise results in cases where data from sensors is not perfect.

Metric	**MUSIC-OMMC**	**Reweighted ADMM-Based Method**
Accuracy	Performs well under Gaussian (ideal) noise, but degrades in impulsive noise conditions	Maintains high accuracy even in harsh noise environments
Low SNR Performance	Performance degrades significantly	More stable and reliable
Handling of Coherent Sources	Fails to resolve coherent sources effectively	Successfully handles coherent sources
Resolution Capability	Limited resolution, especially for closely spaced targets	High resolution with sharp and distinct peaks
Computational Complexity	Low (fast execution)	High (iterative and computationally intensive)
Noise Robustness	Low robustness to impulsive and non-Gaussian noise	High robustness under non-ideal noise conditions

Table 6: Performance Comparison between MUSIC-OMMC and Reweighted ADMM-Based Method

The Table 7 illustrates the comparison of MUSIC using OMMC to the new reweighted ADMM algorithm and how each one provides audio data from the recording system for reproduction of sound. By using OMMC, you can obtain accurate results; however, if there is excess noise or interference, the results will not be very good compared to those obtained using ADMM. Thus, ADMM is superior for processing accurately but takes longer because it needs to repeat a series of processes to produce its final product.

CHAPTER 6
CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This work focuses on developing a practical solution for estimating the Direction of Arrival (DOA) of multiple acoustic sources in underwater environments. In regions such as harbour and coastal areas, the presence of several ships and marine activities leads to overlapping acoustic signals at the receiver. Accurately identifying the direction of each source in such conditions is difficult due to various environmental and system-related factors.

One of the major challenges arises from the way sound propagates underwater. Reflections from the sea surface and seabed create multiple signal paths, which distort the received data. In addition, interference plays a significant role in degrading performance. Signals generated by similar machinery can introduce correlation (coherent interference), while independent activities such as smaller vessels and wave motion contribute to non-coherent interference. Another critical factor is impulsive noise, which does not follow the usual Gaussian distribution and introduces sudden high-energy disturbances in the signal.

Apart from environmental effects, imperfections in the sensing system also impact estimation accuracy. In practice, vector sensors may not be perfectly aligned, resulting in Non-Orthogonal Deviation (NOD). This misalignment affects the array response and leads to errors in direction estimation.

To overcome these challenges, the DOA estimation problem is approached using a sparse signal representation. The angular domain is discretized, and the problem is formulated using an L1-norm (LASSO) framework, which promotes sparsity in the solution. The optimization is solved using a reweighted version of the Alternating Direction Method of Multipliers (ADMM), where the weights are updated iteratively to improve the detection of true signal components while suppressing noise.

A circular Vector Sensor Array configuration is used in this work, providing uniform coverage over all directions and improving the ability to resolve multiple sources. The simulation model includes realistic conditions such as impulsive noise modeled using alpha-stable distributions, along with both coherent and non-coherent interference. The effect of sensor misalignment is also incorporated through the NOD model to better reflect real-world behavior.

The effectiveness of the proposed method is evaluated using performance measures such as Root Mean Square Error (RMSE), Cramér–Rao Lower Bound (CRLB), estimation bias, and resolution capability. The results indicate that the reweighted ADMM-based approach provides improved estimation accuracy compared to conventional techniques, particularly in low Signal-to-Noise Ratio (SNR) scenarios and in the presence of non-Gaussian noise. Additionally, the method is capable of identifying multiple sources without requiring prior information about their number.

In summary, the proposed framework offers a reliable and efficient approach for DOA estimation under practical underwater conditions, combining robustness to noise and interference with manageable computational complexity.

6.2 Future Scope

While the proposed approach shows promising results in simulation, there is still scope for further improvement and extension. One key limitation of the current method is its dependence on angular discretization. Future work can focus on developing a gridless DOA estimation approach, which would eliminate resolution constraints imposed by the predefined angle grid and allow more accurate estimation of closely spaced sources.

Another important direction is the implementation of the algorithm in real-time systems. Since the current framework relies on iterative optimization, it can be computationally intensive. Enhancing the efficiency of the algorithm through faster numerical techniques or implementing it on hardware platforms such as FPGA or GPU can make it more suitable for real-time underwater applications.

At present, the performance of the system has been evaluated only through simulations. A natural extension of this work would be to validate the approach using real underwater acoustic data or experimental setups with actual sensor arrays. This would help in understanding how the method performs under true environmental conditions.

Further improvements can also be achieved by introducing adaptive mechanisms for parameter selection within the ADMM framework. Instead of using fixed parameters, dynamically adjusting them based on signal conditions could improve convergence speed and overall estimation stability.

In addition, the current work focuses mainly on static DOA estimation. Extending the framework to include tracking of moving targets would make it more practical. This can be achieved by integrating tracking techniques such as Kalman filtering or particle filtering to continuously estimate and update target positions over time.

Finally, the model can be extended to account for more complex sensor imperfections, including mutual coupling and gain-phase mismatches. Exploring three-dimensional DOA estimation using advanced array configurations can also broaden the applicability of the system for more sophisticated underwater sensing tasks.

CHAPTER 7
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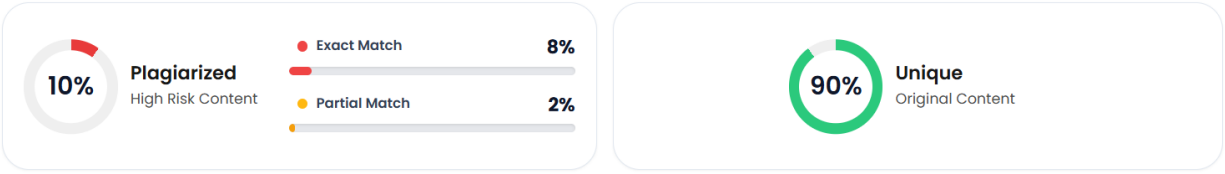
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CHAPTER 8
APPENDICES

8.1 Similarity Check Report

Similarity Check Overview:

- **Total Similarity Index:** 10%



- **Exclusions:** Matches from commonly used technical terminology related to *DOA estimation, Vector Sensor Arrays, sparse optimization, ADMM*, and standard algorithmic descriptions were excluded from the plagiarism count. Properly cited references, equations, and widely accepted signal processing concepts were not considered as similarity.

- **Unique Contributions:**
  The project demonstrates originality through:
  - Development of a multi-ship DOA estimation framework using Circular VSA
  - Implementation of a reweighted L1-based ADMM algorithm for sparse recovery
  - Integration of impulsive noise modelling (alpha-stable) and interference handling
  - Comparative analysis between MUSIC (with OMMC) and proposed ADMM-based method
  - MATLAB-based simulation design, performance evaluation, and result interpretation

- **Result:**
  The similarity analysis confirms that the report is within acceptable academic limits. The work reflects a strong level of originality in methodology, implementation, and analysis, adhering to ethical academic standards.

Remarks:

- All third-party sources, including research papers and standard algorithms, have been properly cited and acknowledged.
- No instances of direct copying or unauthorized use of previously published work were identified.
- The report reflects independent effort in modelling, simulation, algorithm development, and performance evaluation.
- The structure, explanation, and interpretation of results are presented in a clear and original manner.